

Review article

A comprehensive review on solar to thermal energy conversion and storage using phase change materials

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ABSTRACT

Renewable energy plays a pivotal role for mankind in the times of adverse climate change and global warming. However, renewable energy such as solar energy comes with inherent drawbacks of limited or varying availability in terms of time, space and power. Consequently, it will lead to poor performance of numerous solar thermal technologies. To overcome these constraints of solar energy, Thermal Energy Storage (TES) can play a pivotal role in improving performance and feasibility of solar thermal technologies. TES using Phase Change Material (PCM) is one of the effective techniques of charging, storing, and discharging thermal energy as and when required. PCM stores thermal energy in the form of latent heat by undergoing phase change at constant temperature. However, PCM suffers with drawbacks of low thermal conductivity, poor solar to thermal conversion efficiency, and risk of leakage during phase transition. These thermo-physical properties limit the applicability of PCM as a potential TES material. In view of the above facts and findings, this study comprehensively analyses the above mentioned thermo-physical bottlenecks of PCMs. It briefly discusses TES, various materials for TES, PCM, and properties of PCM. In detail, it presents various methods and mechanisms of improving solar to thermal performance and thermal conductivity of the PCM. Additionally, this study presents a robust discussion on techniques of minimizing the leakage of the PCM during phase transformation.

1. Introduction

Renewable energy technologies and its capacity building will play a major role in mitigating the effect of global warming and climate change. Renewable energy, such as solar energy, wind energy, ocean energy, and geothermal energy, plays a crucial role in fulfilling the rising demand for energy in a sustainable way and helps in minimizing emissions caused due to over-exploitation of fossil reserves [1]. Among all the available renewable sources of energy, solar energy is the most abundant source of available energy [2]. Also, it is clean and easily available at almost every location on the earth. However, due to limited availability in terms of time, location, and power, solar thermal technologies such as solar collectors, solar concentrators, solar desalination, and solar driers have poor efficiency and low utilization rates [3], [4]. For example, after sunshine hours, there is no utilization of solar thermal technologies without storage. The intermittent nature of solar energy has caused poor performance of solar thermal technologies [5]. To overcome these drawbacks related to solar energy, researchers have

come up with a solution called Thermal Energy Storage (TES) [6]. TES is the technique of storing heat/cold when available in abundant quantity and utilizing it when it is not available. TES can be classified as sensible heat storage, latent heat storage, and thermochemical energy storage [7]. Among all these three types of TES, latent heat storage (LHS) is widely used because of its various merits such as high energy density and minimum temperature change. LHS is the technique of storing thermal energy due to the phase transition of the storage material. Such material, which undergoes phase transformation while storing and releasing thermal energy, is called Phase Change Material (PCM). In recent years, TES using PCM (solid-liquid) has evolved as one of the prominent techniques for storing excessive heat/cold and utilizing it as and when required [8]. PCM stores and releases thermal energy at an almost constant temperature, have a high energy density, available in wide temperature ranges, and are easily available. For example, rock has a specific heat capacity of 0.9 kJ/kg K, and paraffin wax has a heat storage capacity of 174.4 kJ/kg. It means that to store 174.4 kJ/kg of energy, rock temperature needs to increase by almost 175 K. Also, the storage volume needed to store 1 GJ of energy for rock is 9.9 m³, and

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Nomenclature			
TES	Thermal Energy Storage	EG	Expanded graphite
PCM	Phase Change Material	CPCM	Composite phase change materials
SHS	Sensible heat storage	MNH	Magnesium nitrate hexahydrate
TCES	Thermo-Chemical Energy Storage	CMC	Carboxy methyl cellulose
TCM	Thermochemical materials	NIR	Near Infra-Red
LHS	Latent heat storage	PW	Paraffin wax
CLMS	Carbon layer-modified mesoporous silica	MF	Melamine foam
SS-PCM	Shape stabilized PCM (SS-PCM)	PAA	Polyacrylic acid
DSC	Differential scanning calorimetry	DDA	Dodecanol
PEG	Polyethylene glycol	CSH	Calcium silicate hydrate
GNP	Graphene nanoplatelet	GO	Graphene oxide
CNT	Carbon nanotubes	RGO	Reduced graphene oxide
SWCNTs	Single-wall carbon nanotubes	LSPR	Local surface plasma resonance
MWCNTs	Multi-wall carbon nanotubes	m	Mass of PCM
HNT	Halloysite nanotube	Q	Sensible heat energy
LDPE	Low-density polyethylene	C _p	Specific heat storage
HDPE	High-density polyethylene	ΔT	Change in the temperature
CNs	Carbon Nano-additives	C _{pi}	Specific heat of PCM between T _i and T _m
CNF	Carbon nanofiber	C _{ps}	Specific heat of PCM between T _m and T _f
FAC	Fly ash cenosphere	T _m	Melting temperature of PCM
GA	Graphene aerogel	Δq	Latent heat of fusion
BN	boron nitride	HOMO	Highest occupied molecular orbital
		LUMO	Least unoccupied molecular orbital

that for paraffin is 3.3 m³. Because of these merits, PCMs are widely used for temperature regulation in textiles [9], automobiles [10], [11], batteries [12], [13], [14] buildings [15], [16], aerospace [17], [18], district heating [19], electronics, [20], [21] refrigeration [22] and cold chain supply [23], [24], [25].

Despite the above advantages and applications of PCM in various thermal systems, PCM suffers from four significant drawbacks, i.e., poor thermal conductivity [26], [27], [28] leakage during phase transition [29], and poor solar to thermal conversion [30], [31], [32]. These drawbacks limit PCM applicability in varieties of thermal systems. Poor thermal conductivity reduces the heat transfer from and to the PCM, causing poor charging and discharging of the PCM [33], [34], [35]. The risk of leakage during phase transition leads to loss of PCM from the TES systems. Reduction in quantity due to leakage will hamper the LHS capacity of the TES systems [36]. Poor light-absorbing capacity of the PCM causes low solar-to-thermal conversion efficiency [37] [38], which degrades its thermal performance. Several attempts have been made in past by the scientists to overcome these drawbacks of PCM. Nanoparticles are widely used to improve thermal conductivity of the PCM. Carbon based nanoparticles specifically 1D, 2D, and 3D materials are significantly used by the researchers to improve thermal conductivity of PCM. Additionally, metal-based nanomaterials such as Ag, Ni, Cu, Al, and Ti were also commonly used to enhance heat transfer rate of PCM. Apart from having improved thermal conductivity, nano-enhanced PCM also shows high photo to thermal conversion efficiency [39]. It has been found that PCM loaded with nanoparticles have excellent photothermal conversion efficiency in comparison with bare PCM [40]. Apart from nanoparticles, porous material such as porous carbon, graphite scaffolds, polyurethane foam, and porous clays were also used to improve heat transfer of PCM [41]. These porous materials not only improve thermal conductivity but also act as supporting material to reduce leakage of the PCM. Due to porous structure of the materials, they have larger pore size and PCM settles easily in those pores. Every application of PCM takes advantage of specific properties of PCM. For e.g., solar thermal systems need better solar to thermal conversion along with thermal storage whereas buildings need better heat transfer rate along with thermal storage. Thus, to investigate and understand about various methods, mechanism and materials used to improve thermal

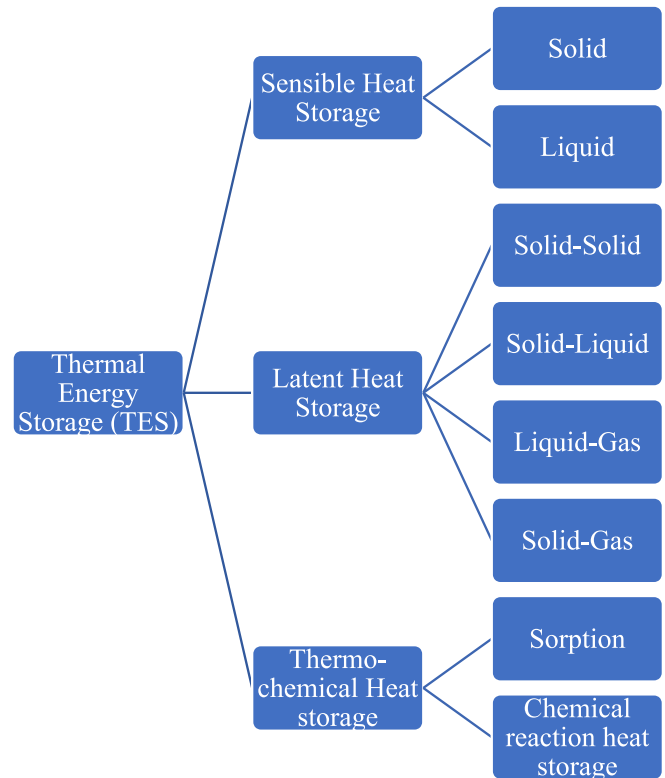


Fig. 1. Classifications of TES methods [43].

performance of the PCM along with anti-leakage characteristics, this comprehensive and updated review study was conducted. Several attempts have been made in the past to review these drawbacks related to PCM. However, none of those studies have investigated all these demerits combined in a single study. Also, there is a scarcity of studies related to detailed analysis of improving solar to thermal conversion and

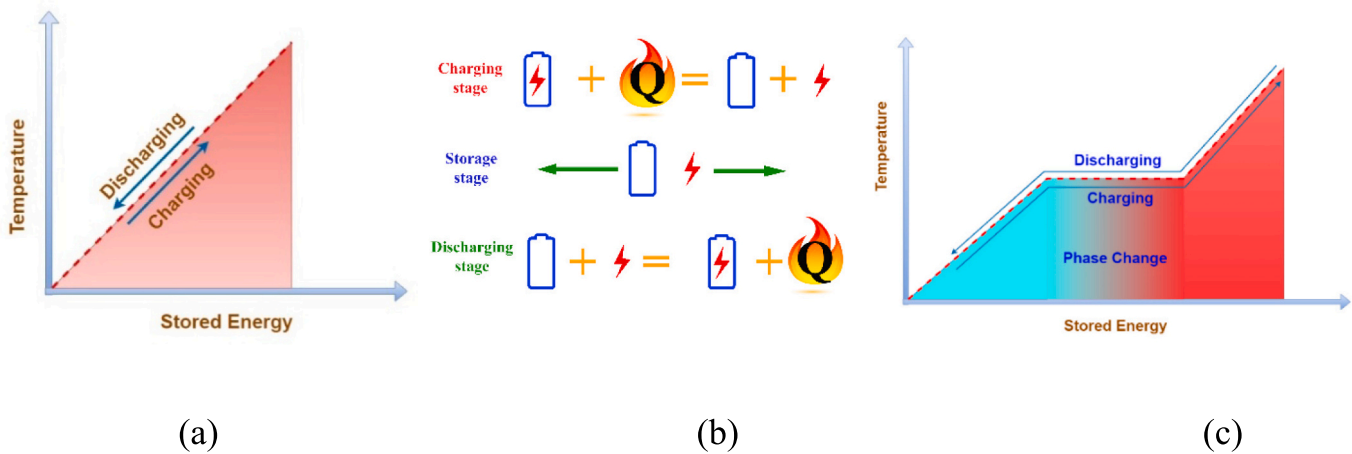


Fig. 2. (a) Sensible heat storage (b) latent heat storage (c) thermochemical heat storage.

Table 1

List properties of different materials for sensible heat storage [42] [46].

Materials	Type	Temperature range (°C)	Specific heat [J/(kg K)]	Density (kg/m ³)
Concrete	Solid	–	880	2240
Marble	Solid	–	800	2600
Rock	Solid	–	879	2560
Steel	Solid	20–70	502	7800
Aluminum	Solid	Up to 160	896	2707
Iron	Solid	Up to 160	452	7900
Brick	Solid	20–70	840	1600
Cement mortar	Solid	–	1000	1800
Sand	Solid	–	800	1555
Gypsum board	Solid	Up to 50	1060	750
Water	Liquid	0–100	4190	1000
Butane	Organic liquid	Up to 118	809	2400
Gravelly earth	Solid	At 20	1840	2050
Glass	Solid	At 20	837	2710
Magnetite	Solid	At 20	752	5177

storage performance of PCM for TES systems. The objective of this study is to conduct a comprehensive review of various methods, materials, and characterization techniques used to overcome the drawbacks of poor thermal conductivity, leakage of PCM, and poor solar to thermal conversion efficiency.

2. Thermal energy storage

TES is one form of energy storage technique which is used by mankind since ancient times. TES is the three-step process i.e. charging, storing, and discharging. TES is one of the prominent techniques of increasing the thermal system's efficiency and helps in overcoming the gap between demand and supply [42]. There are three types of TES i.e. sensible heat storage, thermo-chemical heat storage, and latent heat storage. The detailed classification of TES is shown in Fig. 1.

2.1. Sensible heat storage (SHS)

Sensible heat storage is the simplest and most economical way of storing thermal energy, which stores the heat energy in its sensible heat capacity under the change in temperature, as represented in Fig. 2(a) [44]. For e.g., water can store heat energy either by raising its temperature or changing its phase, but it is known as sensible heat when it stores energy due to rising temperature. The sensible heat energy is

expressed as,

$$Q = m.C_p.\Delta T \quad (1)$$

The sensible heat can be stored in a solid or liquid storage medium. The high-temperature SHS materials are the focus area of the researchers for intermittent solar energy storage applications [45]. The major drawback with SHS is that the amount of heat that can be stored is based on the rise of temperature. To store large energy the temperature rise is also larger. This makes it ineffective for various thermal applications. Commonly used sensible heat substances for building applications are listed in the Table 1.

2.2. Thermo-chemical energy storage (TCES)

Thermochemical materials (TCM) store and release heat by a reversible endothermic/exothermic reaction. TCES technology can be divided into chemical reaction heat storage (CRHS) and sorption heat storage (SOHS). Both technologies offer the following advantages: high TES capacity, low-temperature thermal energy storage, minimal heat losses, and small storage systems [47]. There are three stages for storing thermal energy by TCM charging (reaction/sorption), storage (low temperature-open/close system), and discharging (reversible reaction/sorption), as shown in Fig. 2(b). In a sorption process, the sorbent (usually solid or liquid) and the sorbate (often vapour or gas) are separated during the desorption process using a heat source. The heat energy is released when the sorbent comes back into contact during the sorption process. In this instance, the thermal energy is stored through the chemical potential. The general equation of TCES can be given as



The ability to store energy with practically negligible energy losses is the real benefit of thermochemical energy storage, making it an attractive option for long-term or seasonal energy storage. However, its principal drawbacks include high costs, high complexity, poor heat transfer, and low stability that, with time, reduces storage efficiency. Schmidt et al. [48] used a thermochemical reaction for storing seasonal thermal energy by converting calcium hydroxide into calcium oxide and water. The reaction can be given as $\text{Ca}(\text{OH})_2 \leftrightarrow \text{CaO} + \text{H}_2\text{O}$. The forward reaction needed a temperature of around 450 °C. The addition of zinc or aluminum can enhance the rate of reaction. In a recent development, MgO microparticles with ultrahigh porosity can be used to create environmentally friendly heat storage systems with an efficiency of 90.3 % from the theoretical rate of reaction, which is considerably good [49]. However, further analysis is needed before this material's real-time use for building applications.

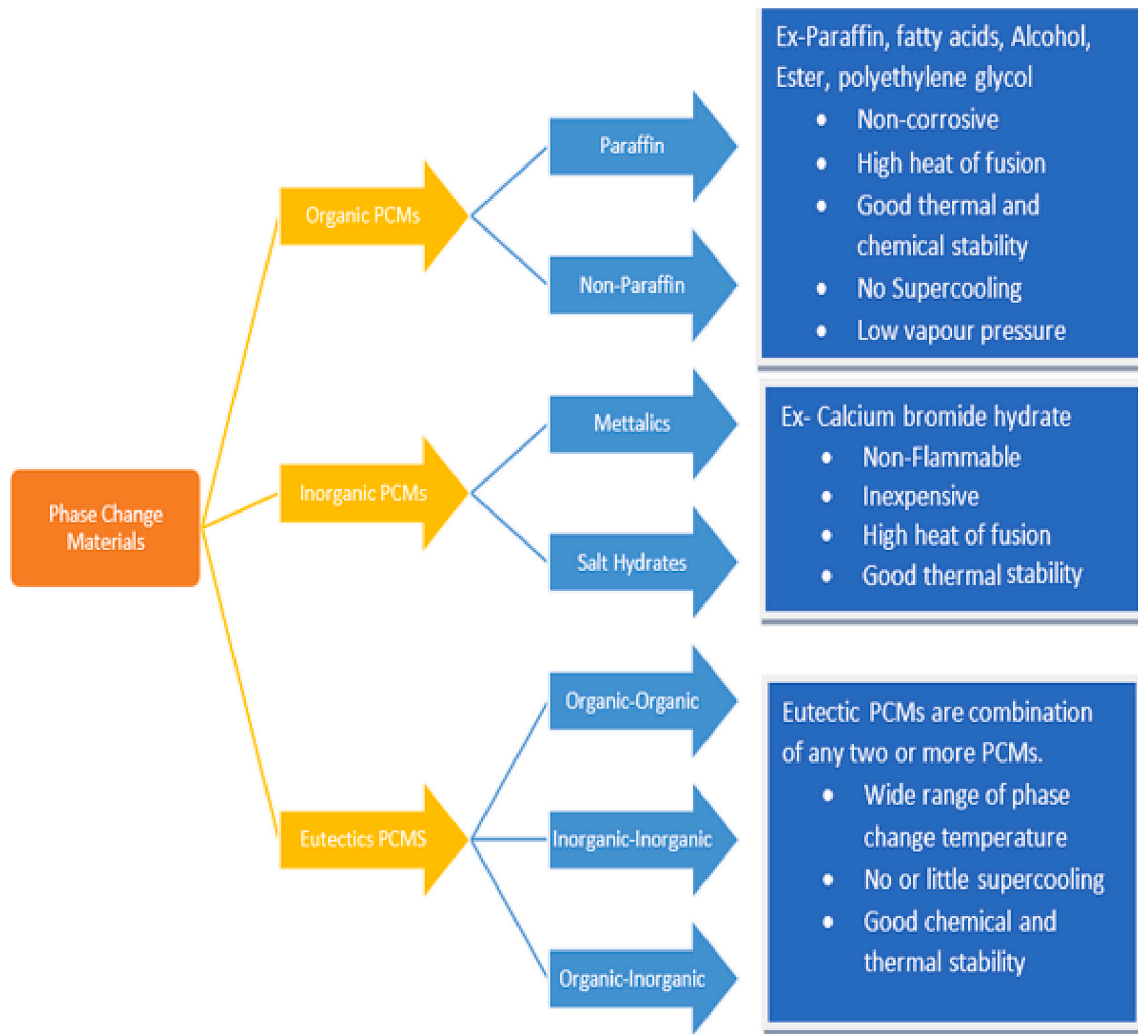


Fig. 3. Classification of PCM with a brief description.

2.3. Latent heat storage (LHS)

The latent heat storage has primarily stored the heat during the phase-change process (almost constant temperature) and is firmly linked to the substance's latent heat as shown in Fig. 2(c). The most effective way to store the latent heat for building applications is by using phase change materials because it has high energy storage density at constant temperature [50]. The heat energy stored by a PCM can be given as,

$$Q = \int_{T_i}^{T_m} mC_{pi} dT + mf\Delta q + \int_{T_m}^{T_f} mC_{ps} dT \quad (2)$$

$$Q = mC_{pi}(T_m - T_i) + mf\Delta q + mC_{ps}(T_f - T_m) \quad (3)$$

PCMs are often used for temperature regulation in various thermal applications due to their advantages like high energy density, low cost, ease of incorporation, availability for a wide range of temperature applications, and isothermal energy storage. Due to the incorporation of PCM, the latent heat capacity of the building increases. In LHS, there are three primary modes in which the phase shift process occurs: solid-solid, liquid-gas, and solid-liquid. Heat is retained in the first scenario by switching between several crystallization forms. Although the latent heat for liquid-gas systems is relatively large, the high-volume variations during phase change provide some issues with storage control. The most common is the solid-liquid PCMs, which have a relatively high melting latent heat and a low volume variation during latent heat exchange

(often less than 10 %) [51]. A suitable PCM with a melting point within the appropriate temperature range, a suitable heat exchange surface, and a suitable container compatible with the PCM are the minimum three components that any latent TES system must have [52].

3. Phase change materials (PCMs)

PCMs can store latent heat when undergoing a solid-to-liquid or liquid-to-gas phase transition. PCMs are energy-saving products with a low energy cost that falls under clean energy sources. PCMs are generally categorised based on their physical transition for heat absorption and desorbing characteristics. PCMs are typically divided into four groups according to their phase transition processes: solid to solid, solid to liquid, gas to solid, and liquid to gas. Solid to liquid PCMs are widely used for thermal applications due to various advantages like high energy density, low volume change, availability, environmentally friendly and low cost. Fig. 3 shows a wide range of classifications and a brief description of PCMs.

Various PCMs are available in different transition temperature ranges and have varied thermo-physical characteristics which affects its TES capacity. Therefore, selection of PCM for any TES system is the crucial factor in improving its thermal performance. The selection of PCMs can be made based on several thermophysical, technical, environmental, and economic considerations. Al-Yasiri et al. [53] conducted a study where five parameters were considered for selecting PCMs:

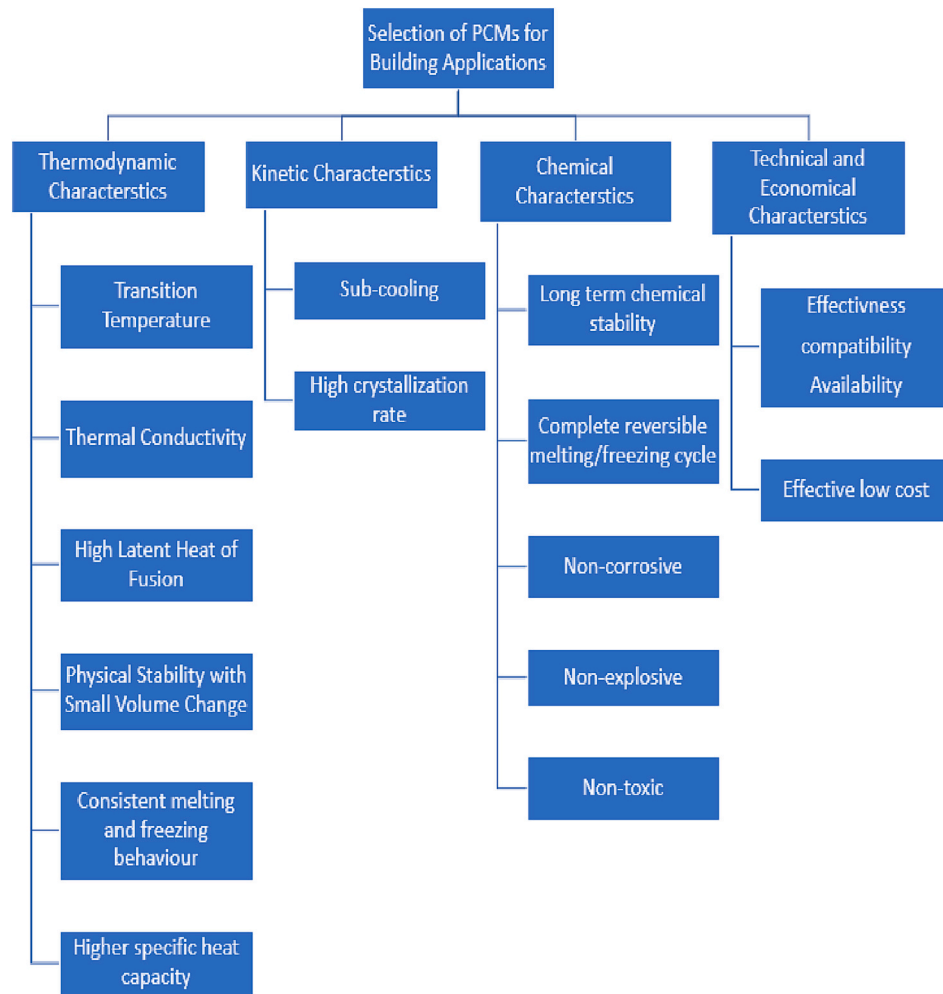


Fig. 4. Selection of PCMs for building applications.

Table 2
Common inorganic PCMs.

PCM chemical formula	PCM types	PCM name	Heat of fusion (J/g)	Melting temperature (°C)	Ref.
Zn (NO ₃) ₂ · 6H ₂ O	Inorganic	Zinc nitrate hexahydrate	147	36.4	[56]
CaCl ₂ · 6H ₂ O	Inorganic	Calcium chloride hexahydrate	171	29.7	[57]
KF · 4H ₂ O	Inorganic	Potassium fluoride tetrahydrate	147	36.4	[58]
Na ₂ HPO ₄ · 12H ₂ O	Inorganic	Sodium orthophosphate dodecahydrate	251	35	[59]
Na ₂ SO ₄ · 10H ₂ O	Inorganic	Sodium sulphate decahydrate (Glauber's Salt)	254	32.4	[60]
CH ₃ COONa · 3H ₂ O	Inorganic	Sodium acetate trihydrate	264	58	[61]
Na ₂ S ₂ O ₃ · 5H ₂ O	Inorganic	Sodium thiosulfate Penta hydrate	200	48	[62]
OM37	Inorganic	OM37	214.18–218.08	32.21–43.47	[63]
Mn (NO ₃) ₂ · 6H ₂ O	Inorganic	Manganese (II) nitrate hexahydrate	128.5	25	[64]
Mn (NO ₃) ₂ · 6H ₂ O + 4 wt% MnCl ₂ · 4H ₂ O	Eutectic	Manganese (II) nitrate hexahydrate & Manganese chloride tetrahydrate	121.1	22.1	
CaCl ₂ · 6H ₂ O · CO(NH ₂) ₂	Eutectic	Calcium chloride hexahydrate-carbamide	149.2	24.18	[65]
Na ₂ S ₂ O ₃ · 5H ₂ O and CH ₃ COONa · 3H ₂ O	Eutectic	Sodium thiosulfate Penta hydrate and sodium acetate trihydrate	211.9	41.58	[66]
KNO ₃ , and NaNO ₃ in ratio of 4:6	Eutectic	Potassium nitrate and sodium nitrate	110	225.7	[67]

phase change temperature, thermal conductivity, latent heat of phase change, compatibility with encapsulation materials, and the material's flammability. The phase change temperature of PCMs is the primary parameter of the selection of PCMs. Based on other literature, it can be concluded that the choice of PCMs for building applications should depend on working temperature range. Working temperature range varies with application such as for buildings it must be 20 °C to 40 °C, for solar water heating it must be around 50 °C to 70 °C, for electronics cooling it must be around 20 °C to 30 °C. The environmental conditions

and the different thermophysical properties can influence the performance of the PCMs [54] [55]. The classification of the selection of PCMs is summarized in Fig. 4. The use of eutectic PCMs for TES has increased as eutectic PCMs can be prepared at any desired transition temperatures and have properties similar to that of the starting components.

Some commonly used organic, inorganic, and eutectic PCMs, are summarized in Tables 2 and 3. Commonly available PCMs for the different temperature ranges are shown in Fig. 5, and the commercial grade PCMs useful for various thermal applications are summarized in

Table 3
List of organic and eutectic PCM.

Ref.	PCM types	Material	Latent heat of storage (J/g)	Temperature range (°C)	Remarks
[68]	Organic	Paraffin	~200	20-60	–
[69]	Organic	Lauric acid	146	37	–
[70]	Organic	Stearic acid	194	53.1	–
[71]	Organic	n-Eicosane	250.41	28.87-37.27	–
[72]	Organic	One dodecanol	200	26	–
[73]	Organic	One tetradecanol	205	38	–
[74]	Organic	Cetane	231	18	–
[75]	Organic	Hexadecane	236	18	–
	Organic	Heptadecane	214	22	–
[76]	Organic	Octadecane	215.6	30.33	–
[77]	Organic	Isopropyl-palmitate	112.2-113.6	9-13	–
[78]	Organic	n-Nonadecane	243.5	27-32	–
[79]	Organic	Paraffin RT18	134	18	–
[80]	Organic	n-Octadecane/melamine formaldehyde	122.24	26.01	–
[76]	Organic	Octadecane/silicone	103.8	34.4-37.8	–
[81]	Organic	Nano-encapsulated PEG600 into a silica shell	66.24	20-21	The heat transfer rate increased
[82]	Organic	Lauric acid, nano-Cu and attapulgite	87.1	42.1	0.46 W/(m-K)
[83]	Organic	50 % paraffin wax & 50 % high-density polyethylene	53	48-51	0.35 to 0.385 W/m K
[84]	Organic	Octadecanoic acid, octadecanol and expanded graphite	135.2-143.5	45.55-47	3.38 W/(m-K)
[85]	Eutectic	Capric and lauric acid	143	21	–
[86]	Eutectic	Polyethylene glycol/silica (PEG@SiO ₂)-coated eutectic hydrated salt (EHS)/fly ash cenosphere (FAC) composite	129.7	15.2 °C and 23.5 °C	Energy saving around 0.248 kWh/kg
[87]	Eutectic	Lauric acid, myristic acid, and palmitic acid	143	33.5	–
[88]	Eutectic	Myristic acid and tetradecyl alcohol	198	28.5-42.5	–
[89]	Eutectic	Lauric acid-1 and hexadecanol	41.8	200	–

Table 4.

3.1. Incorporation of PCMs

Phase change materials (PCMs) can be incorporated into building materials in various ways to improve energy efficiency and thermal comfort, summarized in Fig. 6. Each method has its advantages and disadvantages depending on the type of building, application, and material. It is essential to consider the specific requirements of the building, including cost, durability, and energy efficiency, when selecting the appropriate method for incorporating the PCMs.

The challenges associated with PCMs as potential TES materials are low thermal conductivity, leakage, volume change during phase transition, flammability, supercooling, corrosiveness, and poor stability [95]. However, leakage and thermal conductivity are the most challenging issue to be addressed. When PCM gets embedded in any thermal

system, TES must be leakproof. The leakage of PCM during phase transition from the TES systems will degrade the TES capacity of the system and, consequently, inversely affects its thermal performance. The leakage issue of PCMs was addressed using different incorporation techniques like encapsulation and shape stabilization [96]. The use of incorporation techniques must ensure minimal leakage during the melting of the PCM [97]. PCMs incorporated into thermal systems using different techniques are briefed in the Fig. 7.

3.1.1. Direct incorporation

Direct incorporation technique is primarily used in building envelopes. In direct incorporation, the PCM in powder or liquid form is added directly to the construction material, such as gypsum mortar, cement mortar, and concrete [99]. There is a particular disadvantage with direct incorporation, such as leakage causing material incompatibility and increasing the risk of fire (for flammable PCMs) along with loss of PCM. Furthermore, because the PCM is added to the mixture in a liquid state, the water content ratio is reduced, which weakens the mechanical properties of constructed elements at high temperatures [100]. The durability and bonding strength of aggregates and binder paste of building materials were also reduced [101]. [102] conducted an experimental investigation on the thermal performance of walls with voids filled with cellulose insulation augmented with PCM. The findings revealed that individual walls had daily average peak heat flux reductions of 25.4 % and hourly average peak heat flux reductions of 20.1 % for all the walls combined. In another work, Kontoleon et al. [103] investigated the thermal and mechanical properties of the reinforced concrete wall covered with PCM-enhanced plastering by directly mixing the obtained capsules with the mixture of cement CEM II 32.5 and natural sand. The result showed that the interior thermal comfort increased and heat flux through the wall was reduced, and the interval between the peak temperatures inside and outside was increased. Tian et al. [104] evaluated the effectiveness of a wall coated with plastering mortar that included 50 % ceramsite PCM to a control wall coated with pure mortar. Results showed that the inner surface of the wall had a 55 % reduction in temperature variations, demonstrating the usefulness of PCM-enhanced mortar in achieving interior thermal comfort in the summer.

3.1.2. Impregnation of PCM

In this method, the supporting materials are dipped in the liquid PCM, where the material absorbs the liquid through capillary action, an illustrative image shown in Figs. 8 & 9 [105]. The impregnation process has three stages. First, air and water are removed from the base material with the help of a vacuum pump. Then the base materials are dipped into the liquid PCM under a vacuum. Finally, the PCM-dipped material is mixed with the concrete [106]. The main disadvantages of this method are leakage, incompatibility with base material, and corrosion of reinforced steel when incorporated with concrete elements, which reduces its service life [107]. Qu et al. [108] created a variety of samples with various paraffin and fume silica in different mass ratios. It was suggested to use a raw material mass ratio of 45:55 for paraffin and fumed silica, respectively. The high capillary pressure of the nanopores and the van der Waals interaction between the methyl groups rendered the composite stable at this proportion. In another work, Chen et al. [109] investigated carbon layer-modified mesoporous silica (CLMS).

3.1.3. Encapsulation of PCMs

Encapsulation of PCMs refers to the process of enclosing the PCM within a shell or capsule. Due to leakage, environmental reactivity, and toxicity issues during phase change, PCMs cannot be directly incorporated into the thermal envelope. Encapsulation provides a coating or covering over the PCM. Encapsulation is the preferred method to incorporate a PCM in thermal systems because it effectively prevents PCM leakage, which can result in a loss of efficiency and performance. Encapsulation can also help improve the PCM's stability and durability,

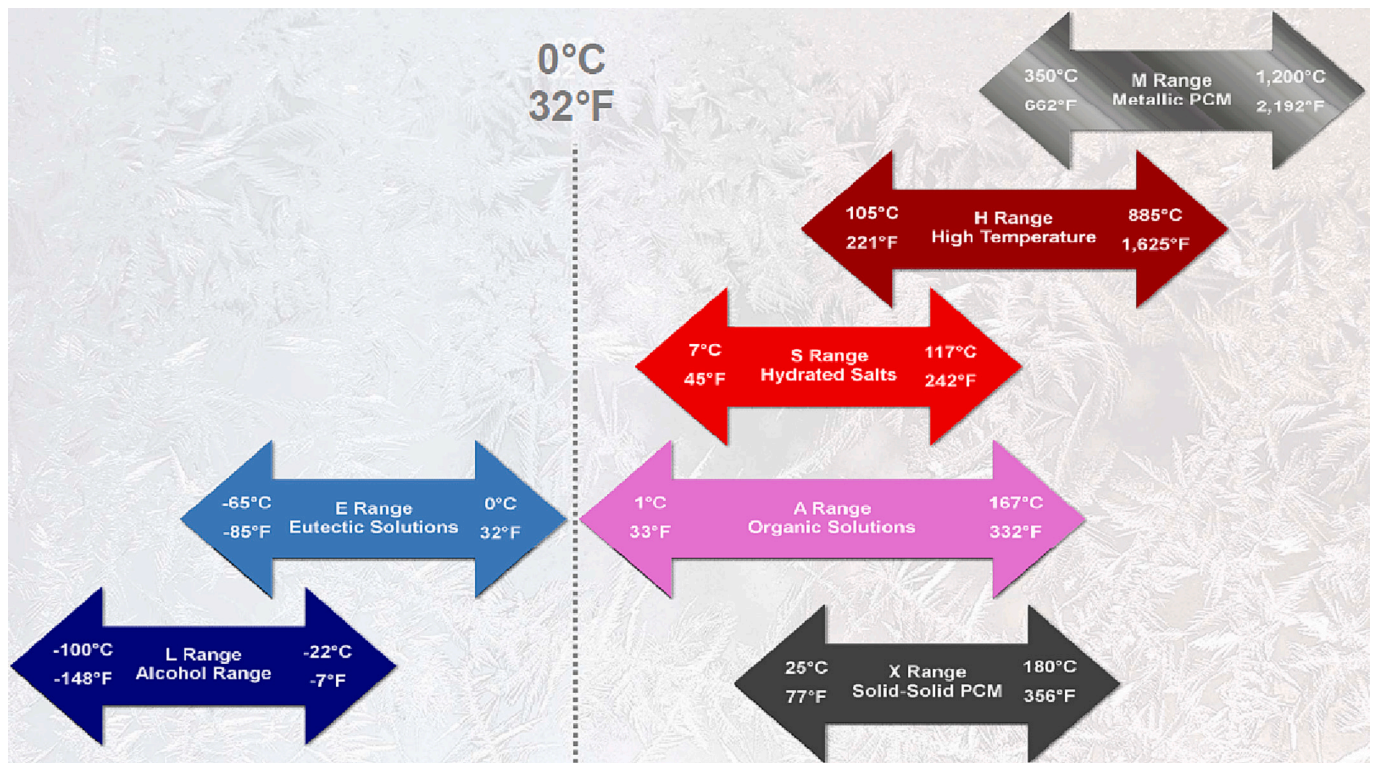


Fig. 5. Temperature range of different types of commercial-grade PCMs [45].

Table 4

Commercial grade PCMs suitable for building applications.

Ref.	Name of PCM	Melting temperature (°C)	Thermal conductivity (W/m K)	Density (kg/m ³) (solid)	Latent heat of storage (J/g)	Type	Maximum operating temperature (°C)
[90]	A21	21	0.22	770	160	Organic	200
	A22	22	0.18	785	160	Organic	200
	A23	23	0.18	785	155	Organic	200
	A24	24	0.18	790	155	Organic	200
	A25	25	0.18	785	150	Organic	200
	A27	27	0.22	768	250	Organic	200
	A29	29	0.22	810	225	Organic	200
	A32	32	0.21	820	240	Organic	250
[91]	PCM20	18-20	0.14-0.23	680-950	171	Organic	–
	PCM21	19-21	0.14-0.23	730-936	186	Organic	–
	PCM22	20-22	0.14-0.24	780-923	208	Organic	–
	PCM23	21-23	0.15-0.25	830-910	227	Organic	–
[92]	PCM24	22-24	0.15-0.25	840-930	207	Organic	–
[93]	RT21	20-23	–	–	190	Paraffin	–
	RT26	24-26	–	–	232	Paraffin	–
	S21	21	0.54	1530	95	Hydrated salt	140
	S22	22	0.54	1530	93	Hydrated salt	140
	S23	23	0.54	1530	86	Hydrated salt	140
	Climsel C23	23	–	–	148	Hydrated salt	–
	S24	24	0.54	1530	77	Hydrated salt	140
	S25	25	0.54	1530	75	Hydrated salt	140
	S27	27	0.54	1530	80	Hydrated salt	140
	S32	32	0.51	1460	95	Hydrated salt	140
	OM37	37	0.13-0.16	860	179	Inorganic	120

making it more resistant to degradation over time. This method also increases the heat transfer area and, as a result, the thermal conductivity of PCM to use its storage capacity fully. There are various methods for

encapsulating PCMs, including spray drying, emulsion polymerization, and coacervation. These methods involve creating a shell or coating around the PCM, typically using a polymer material, to create a stable

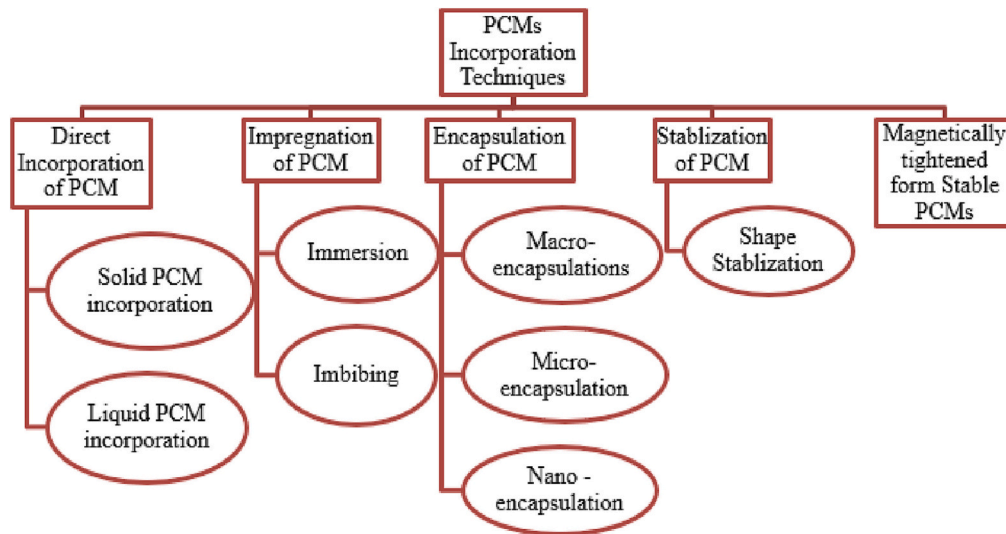


Fig. 6. Classification of different PCMs incorporation techniques.

and contained unit. Three encapsulation types are commonly used, macro, micro, and nano-encapsulation. The scale range of capsules for different encapsulations is shown in Fig. 10. The choice of encapsulation method depends on the specific properties of the PCM, the desired encapsulation material, and the intended application. Encapsulation of PCMs is an important technology for a wide range of applications, including building and construction, transportation, and energy storage. By encapsulating PCMs, it is possible to create highly efficient and effective TES systems that can be used to reduce energy consumption and improve overall system performance.

3.1.3.1. Macroencapsulation. Macroencapsulation of the PCM is the process of encapsulating the PCM in the shell material having a size of more than 5 mm [112]. Macroencapsulation is mainly preferred for PCM encapsulation due to its cost-effectiveness. When macroencapsulated PCM is directly incorporated into the building envelope, it enhances the structure's ability to store latent heat, which enhances the building's indoor temperature profile. The reduction in peak temperature, thermal amplitude, and time lag may indicate improved interior thermal performance. PCM can be macroencapsulated in different shapes, like shells, tubes, channels, and thin plates. Due to the variety of shapes and sizes, it can be easily incorporated into the structure of a building. In the development of macroencapsulation [113], coconut oil as PCM, enclosed in a thin plastic bottle, symbolizing a decorative PCM panel fastened to the ceiling. According to the findings, the time needed to melt the PCM totally may be reduced by 20 % with a 50 % increase in the exposed surface area to volume ratio. Memon et al. [114] prepared macrocapsules of paraffin by incorporating it into lightweight aggregates (LWA) by vacuum incorporation. The results showed that the maximum percentage of paraffin absorbed by LWA was 70 %, and the enhanced thermal conductivity of PCM by using 15 % of graphite powder was 162 % more than epoxy. Dong et al. [115] used octadecane paraffin macroencapsulated in a hollow steel ball, carrying 80.3 % of PCM by mass. The latent heat capacity of capsules was 200.5 J/g. The compressive strength of concrete decreases with the increasing content of macroencapsulated PCM in concrete. Similarly, Cui et al. [116] used a hollow steel ball for macroencapsulation of octadecane with a metal clamp, as shown in Fig. 11. The clamp improves the compressive strength of the composite. The latent heat of macroencapsulated PCM was 153.1 J/g.

Vicente et al. [117] prepared three different samples M_1 – masonry wall (without PCM as reference), M_2 – masonry wall with PCM; M_3 – masonry wall with PCM and insulation material, M_2 & M_3 with hollow-

fired clay brick dimensions $30 \times 20 \times 15$ cm, as shown in Fig. 12. Compared to M_1 , it was discovered that M_2 and M_3 's thermal amplitude had decreased by 50 % and 80 %, respectively. Moreover, M_2 and M_3 reached peak temperatures 3 h later than expected. In recent years micro and nano-encapsulation have been a critical area of research to improve thermal conductivity and make it cost-effective for building applications. Rathore et al. [118] investigated the macroencapsulated PCM (OM37) in aluminum pipes, and the result showed that the thermal amplitude was reduced by 40.67–59.79 %. The peak temperature of walls was reduced by 7.19–9.81 % with a time lag of 60–120 min. Baruah et al. [119] developed a numerical model to simulate the PCM-metal foam hybrid system, and the results showed that the melting and solidification of the hybrid system were enhanced. Lee et al. [120] evaluated the heat flux reduction by optimizing the location of PCM in the south and west walls, and it was 2.54 cm and 1.27 cm from the wall-board, respectively. The heat flux reduces by 51.3 % and 29.7 % for the south and west wall, respectively. Jin et al. [121] investigated PCM thermal shield for peak heat flux reduction, and the results showed that the heat flux was reduced by 11 % when the thermal shield was placed in the inward-most location. Kong et al. [122] investigated a passive cooling of the building by mathematical modelling and numerical simulation. The results showed the potential energy saving as heat flux reduced for different locations and combinations of PCM. Jin et al. [123] placed the PCM in the wall at a different position for potential reduction in heat flux. The ideal PCM position was $1/5$ L from the interior surface of the wall (L being the thickness of the cavity). Lee et al. [124] developed a testing model called plug-and-play walls (PPW) for faster installation. The findings showed that the south- and west-facing walls have an average daily reduction in heat transfer of 27.4 % and 10.5 %, respectively. Silva et al. [125] investigate a wall specimen containing macroencapsulated PCM bricks and bricks without PCM. The findings show that the wall specimen made with macroencapsulated PCM has a reduction in thermal amplitude of 5–10 °C and a time delay of approximately 3 h. Kong et al. [126] developed three different rooms for testing using macrocapsules and perforated bricks. The first test room is constructed without PCM, the second is constructed with macroencapsulated capric acid on the exterior, and the third is constructed with macroencapsulated 1-dodecanol on the interior. The findings showed that the test room with PCM on the interior surface had the most significant drop in peak temperature. The macroencapsulated also improves the time delay by 2.1 h and 3 h for the outside surface PCM test room and inside surface PCM test room, respectively.

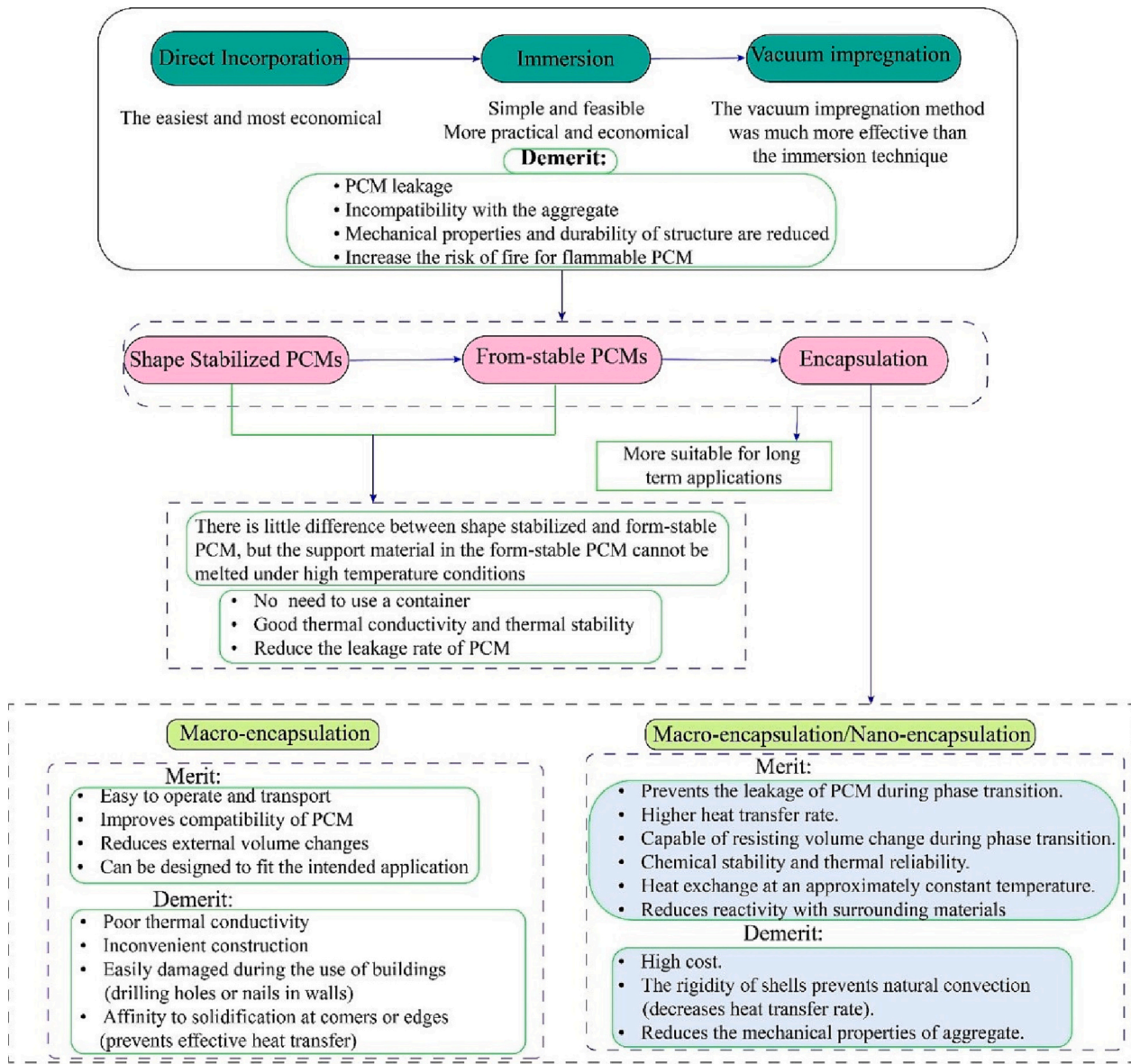


Fig. 7. Brief description of different types of incorporation techniques of PCMs [98].

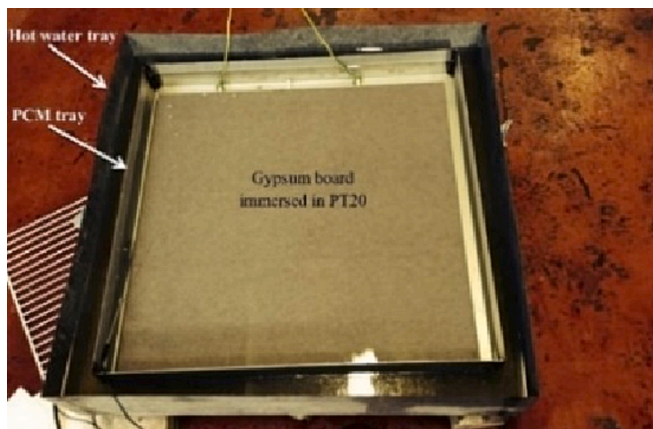


Fig. 8. Gypsum board immersed in PCM for impregnation [105].

3.1.3.2. Microencapsulation. Microencapsulation involves micron size particles, such as solids, liquids, or gases, enclosed in a protective shell or coating. Microencapsulation uses a film-forming material to coat a solid or liquid and form 0.5–5000 μm particles [127]. By encapsulating PCM in a protective shell, the material's TES and release properties can be preserved and enhanced. The protective coating prevents the PCM from leaking or evaporating and provides a controlled release mechanism for the stored energy. An illustrative shell with a core of micro-encapsulated PCM is shown in Fig. 13.

In the microencapsulation technique, individual particles or droplets are coated with a continuous layer to create micro to millimetre-sized capsules. A PCM is the core of microencapsulated phase change materials (MPCM), while a polymer or inorganic shell is the PCM container [129]. Some of the familiar shell materials for microencapsulation of PCM are shown in Fig. 14.

Microencapsulated PCM is composed in powder form or spread in the fluid. It can considerably improve the heat transfer efficiency of a carrier. According to the synthesis mechanism, microencapsulation techniques may be divided into three groups: physical, chemical, and physical-chemical. The detailed classification with brief descriptions is

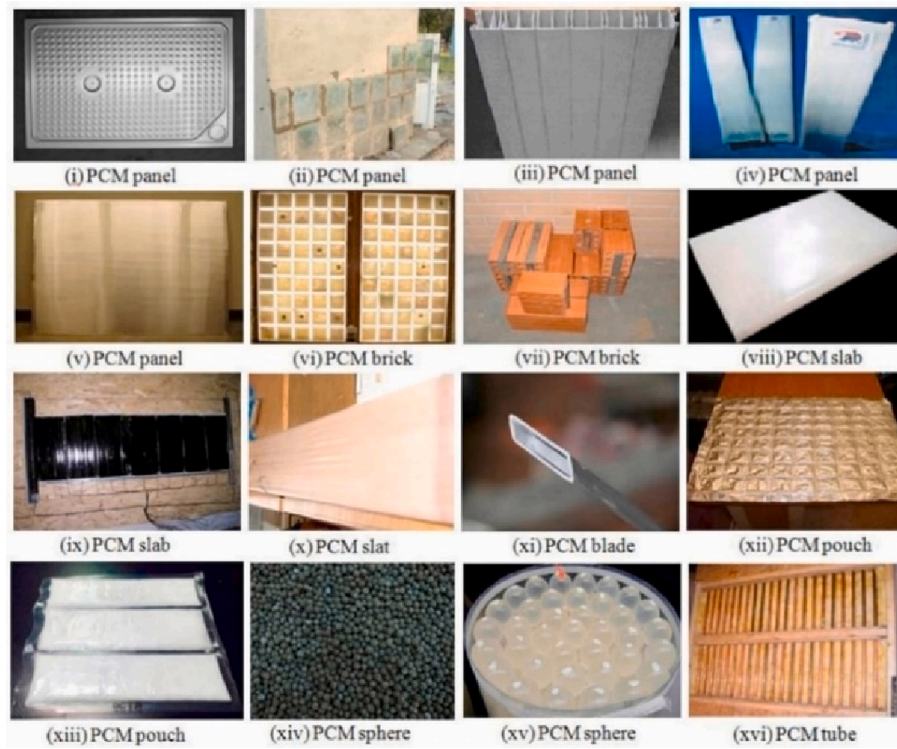


Fig. 9. Commonly used macro-encapsulation for the different building envelopes [110].

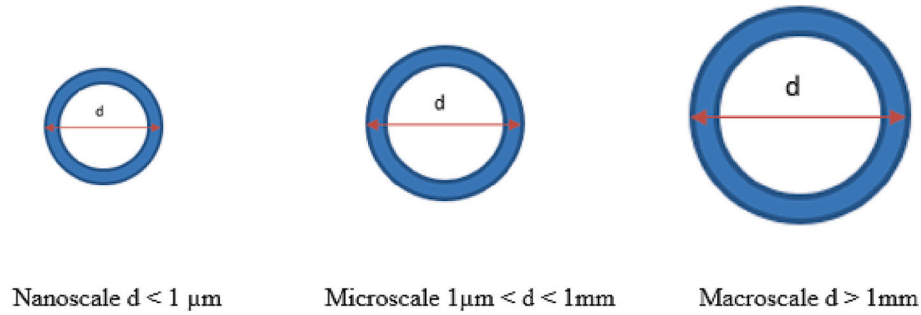


Fig. 10. Nano, micro, and macro range of scale for encapsulation [111].

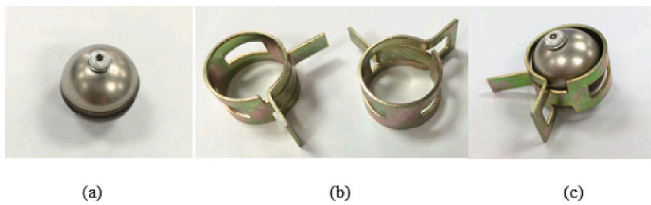


Fig. 11. (a) Hollow steel ball (b) clamp (c) hollow steel ball attached with metal clamp.

shown in Fig. 15.

However, a few characteristics of microencapsulated PCM produced by these techniques are essential, like encapsulation efficiency, ratio, and particle size. Microencapsulation efficiency measures how much target material is successfully encapsulated within the microcapsule. It is expressed as the percentage of the total amount of material added that is actually encapsulated within the microcapsule. The encapsulation efficiency of microencapsulated PCM can also be defined as the ratio of the mass of the core to the mass of the shell, which reflects the core

contents in microcapsules. However, it is difficult to directly measure the mass of the core and shell for a single microcapsule. Therefore, encapsulation efficiency is usually evaluated by calculating the ratio of the latent heat of PCM microcapsules to that of pure PCMs. Mathematically it can be given as an equation, where $\Delta H_{\text{microcapsules}}$ is the latent heat of microcapsule containing PCM and ΔH_{PCMs} is the latent heat of pure PCMs. $\Delta H_{\text{m,microcapsules}}$ and $\Delta H_{\text{c,microcapsules}}$ are latent heat of melting and crystallization of microcapsules respectively; $\Delta H_{\text{m,PCMs}}$ and $\Delta H_{\text{c,PCMs}}$ are latent heat of melting and crystallization of pure PCMs.

$$\text{Encapsulation efficiency} = \frac{\Delta H_{\text{microcapsules}}}{\Delta H_{\text{PCMs}}} \times 100\% \quad (4)$$

Or,

$$\text{Encapsulation efficiency} = \frac{\Delta H_{\text{m,microcapsules}} + \Delta H_{\text{c,microcapsules}}}{\Delta H_{\text{m,PCMs}} + \Delta H_{\text{c,PCMs}}} \times 100\% \quad (5)$$

Microcapsules can be used in a variety of applications, including thermal energy storage, textiles, building materials, and electronics, as shown in Fig. 16. One of the most popular microencapsulated PCM for the building application was developed by Microtek and branded as micronal PCM. The different properties and encapsulation methods of

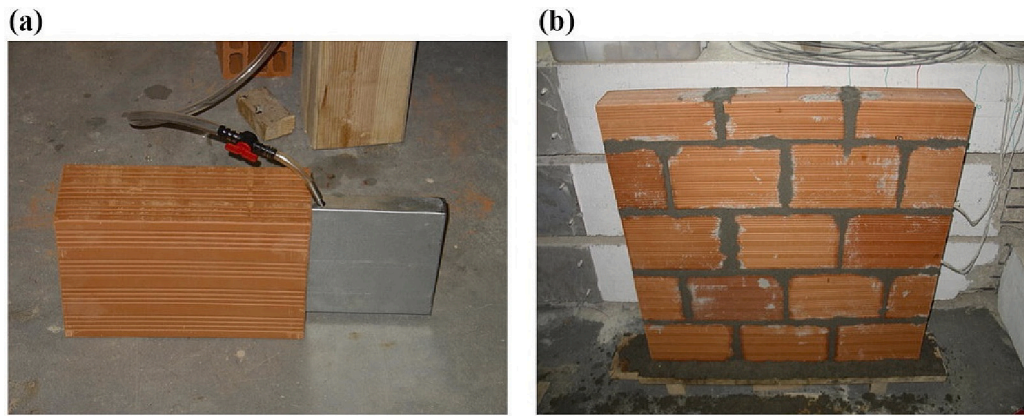


Fig. 12. Experimental setup of hollow-fired clay brick [117].

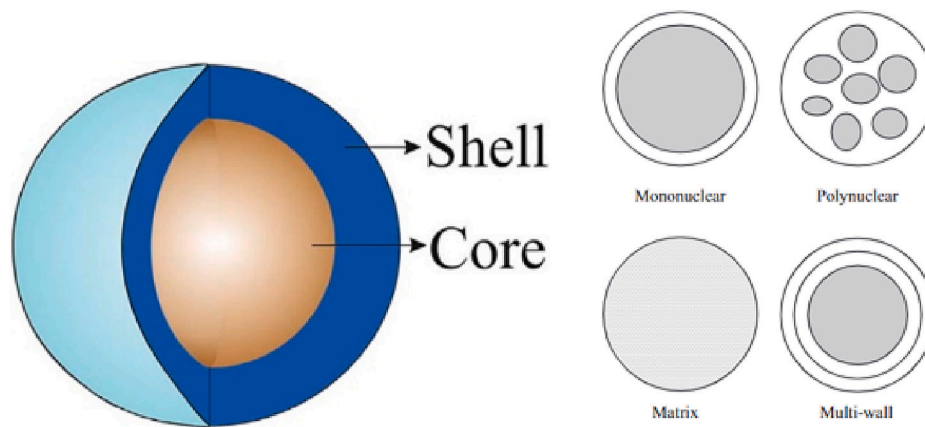


Fig. 13. Shell of microencapsulated PCM [128].

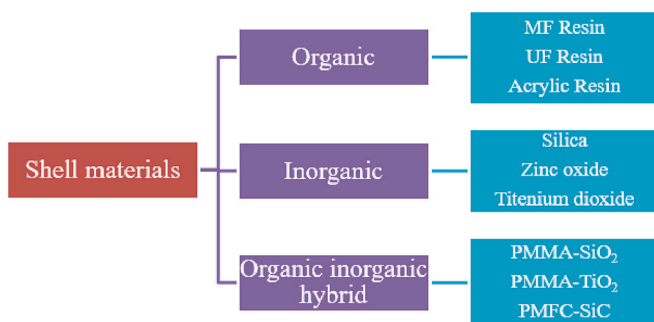


Fig. 14. Different types of shell materials for microencapsulation [130].

micronal PCM are summarized in Schmidt [132]. Arce et al. [133] used the Micronal® PCM with a theoretical melting point of 26 °C and latent heat value of 110 J/g in three panels (the South, West, and roof walls) mixed with the concrete. The cubicles have external measurements of 2.64 m × 2.64 m × 2.52 m, and the panels are 0.12 m thick. The placement of the windows is as follows: a door on the north wall, four windows of 0.75 m × 0.4 m at the south wall, and one measuring 1.7 m × 0.6 m at the east and west walls. During the summer, south wall windows can be opened for night time ventilation. The result showed that Peak temperatures were reduced by around 6 %. Additionally, PCM continued to operate for at least 4 % more hours, and comfort time in cubicles with awnings increased by at least 10 %. However, high outside temperatures and solar radiation were still a factor, as PCM did not

perform daily complete phase change cycles as intended. The micronal PCM used by Castell et al. [134] with concrete to validate a methodology for cases with heat gains and identify the intensity of thermal discomfort as the most significant indicator. Furthermore, Castellón et al. [135] also used micronal PCM in sandwich panels to demonstrate the thermal inertia's viability and lower the final building's energy requirement. Eddahak-Ouni et al. [136] investigated the micronal PCM with concrete, and findings showed that adding PCMs increased the PCM concrete's ability to store heat. A vital link was found between the experience and the homogenization technique for predicting thermal conductivity. Joulin et al. [137] incorporated cement mortars with micronal PCM DS 5001 X. Due to the addition of micro PCMs in mortar, the apparent specific heat capacity is 1.35 times greater. The energy stored and released by the PCM-M during the phase change temperature range is 41 % greater than that stored and released by the traditional mortar over the same temperature range. Giro-Paloma et al. [138] analyzed the mechanical properties of micronal PCM, and the elastic response of the particles at various temperatures was evaluated using atomic force microscopy in nanoindentation mode. The key finding is that MPCM stiffness is temperature and particle-size dependent. Cabeza et al. [139] incorporated micronal PCM in concrete cubicles. The study's findings demonstrate how encapsulating PCMs in concrete improves thermal inertia and lowers interior temperatures compared to standard concrete without PCMs. Vieira et al. [140] investigated micronal DS 5008 X PCM incorporated with nanoparticles of titania and superabsorbent polymer. The PCM-containing mortars could attenuate and postpone the temperature variations, while the superabsorbent particles produced a larger hygroscopic capacity. In the meantime, 40–45 % of NOx was photo-catalytically degraded by nanoparticles. However, the

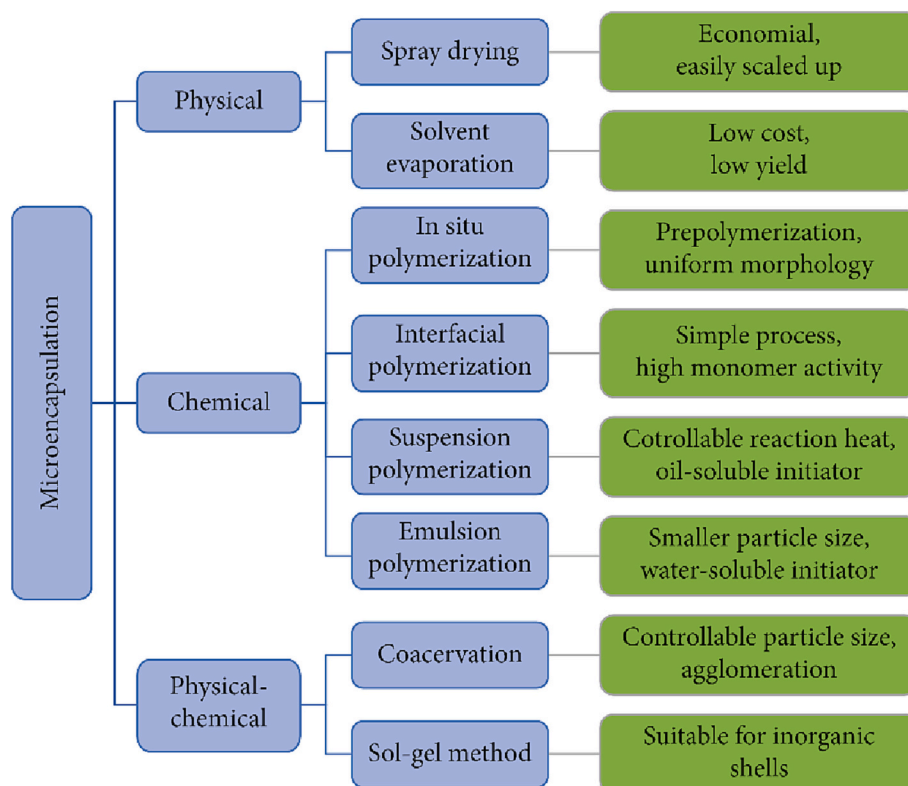


Fig. 15. Brief description of different methods of microencapsulation technique [131].

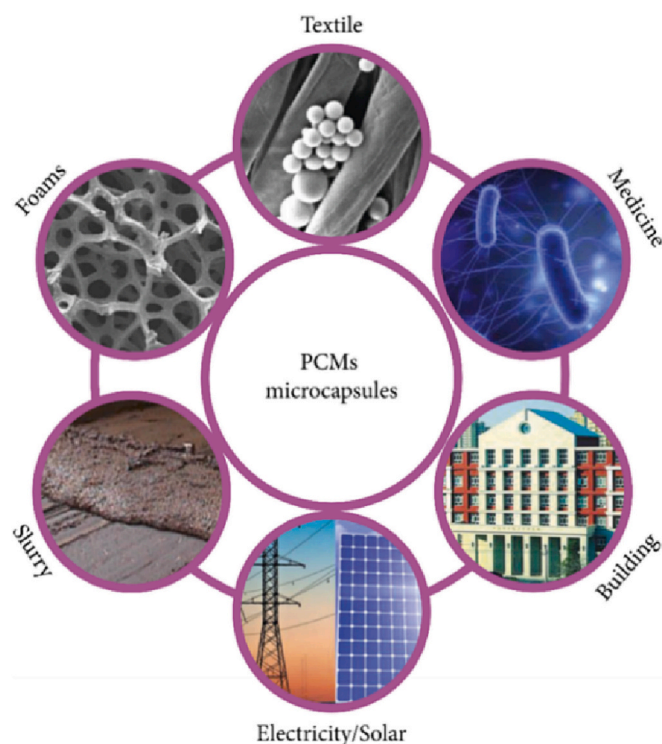


Fig. 16. Different applications of microencapsulation of PCM [131].

presence of PCM particles in both cases—photocatalytic degradation and hygroscopic capacity attenuated the performance.

The brief about the microencapsulation PCM investigated in literature is summarized in Table 5 with shell materials and different

properties like melting temperature particle size, latent heat, and encapsulation efficiency.

3.1.4. Shape stabilization of PCM (SS-PCM)

The shape stabilized PCM contained within a carrier matrix. This method is promising because it has several advantages leakage proof, improved thermal conductivity, a high specific heat, and the ability to maintain the shape over many phase transition cycles. So far, six techniques for SS-PCM encapsulation have been discovered: melt infiltration, cold compression, uniform mixing, vacuum impregnation, cold infiltration, and sol-gel [161]. This method is mainly recommended for organic PCM to enhance thermal conductivity and reduce the leakage issue of organic PCM [162]. Mainly two types of supporting materials are included in the shape-stabilization method for the PCM. These materials include high-conductivity nanoparticles for improving thermal properties and porous material to avoid PCM leakage. Singh et al. [163] used SS-PCM composite with a eutectic mixture of lauric acid and myristic acid, and the result showed that the SS-PCM-39 % was stable and arrested the leakage of eutectic PCM. Huang et al. [164] reviewed the SS-PCM in which three categories were used in the study to classify recent applications: metal foams, porous carbon, porous oxide, and other porous supports. A detailed description is shown in Fig. 17.

Baskar et al. [165] investigated a form stable PCM-based roof system. PCM was a eutectic mixture of lauric and palmitic acid, and nano-silica was used as a thermal conductivity enhancer. The four hut prototypes of size 1.2 m × 1.2 m × 1.2 m were built to study the inside temperature of various roof systems, as shown in the Figs. 18 and 19. The performance of SS-PCM showed a reduction in indoor temperature up to 8 °C–10 °C concerning the conventional model. Another work by Liu et al. [166] investigated an SS-PCM based on hyper cross-linked polymers. The thermal conductivity of SS-PCM increased by 17 % concerning pure PCM. Boussaba et al. [167] prepared a composite PCM based on coconut fat and showed the potential for enhancing building envelope thermal performance.

Table 5
Microencapsulation PCM investigated in the literature for different applications.

PCM	Shell material	Particle size (μm)	Melting temperature (°C)	Latent heat (J/g)	Encapsulation efficiency (%)	Ref.
Pure Temp 6	Cross-linked polymethyl methacrylate	139-317	8.2	131.1	77.2	[141]
75 % SA + 25 % CA	Melamine formaldehyde	10.41	34.5	103.9	85.3	[142]
RT-21	Poly-methyl-methacrylate	6.5	25–30	–	85.8	[143]
Pentadecane	Poly melamine-urea-formaldehyde	2	8.1–8.7	84.5–88.2	48	[144]
Caprylic acid	Polyethyl acrylate	0.64	13.3	77.3	–	[145]
Octanoic acid	Urea formaldehyde and melamine formaldehyde	0.2–1.5	13.9	93.9	59.3	[146]
Paraffin RT21	Polyurea	2.4	27.5	92.5	–	[147]
n-Tetradecane	Polymethyl methacrylate -co-methacrylic acid	1–15	32	127.3	66.72	[148]
C17-C24	Polymethylmethacrylate	1.16–6.42	20.43	37.5	–	[149]
C20-C24	Polymethylmethacrylate	1.16–6.42	35.88	169.26	–	[149]
n-Octadecane	Melamine-formaldehyde	20–71	26.91	146.5	92 %	[150]
Paraffin	Acetylated nanocellulose	50 – 300	55	173	80.2	[151]
Palmitic acid	TiO ₂	0.2–0.4	61.7	63.3	30.4	[152]
Hexadecane	Soy glycinin	101.7	–	–	65	[153]
n-Octadecane	Polyurea microcapsules	35 - 500	27–29	214–216	100	[154]
n-Tetradecane	Calcium carbonate	5	5–9	58.54	25.86	[155]
n-Octadecane	n-Octadecyl methacrylate methacrylic acid copolymer	1.6–1.7	26.5	93	12–21	[156]
n-Octadecane	Silica	7–16	26.9	123	57.5	[157]
n-Octadecane	Melamine formaldehyde	2.2	40.6	144	–	[158]
Micronal® PCM	Acrylate polymer	2–20	26	110	–	[133]
Micronal® DS 5001X	Acrylate polymer	5	27.8	99	–	[159]
Micronal® DS5001	Acrylate polymer		10–30	145	–	[160]

4. Different techniques for enhancing the thermal conductivity of PCMs

4.1. Nanoparticle enhanced PCMs

Nanoparticles were used to enhance the thermal conductivity of organic PCMs because of the high surface-to-volume ratio. The thermal conductivity improvement of shape-stabilized PCM loaded with nanoparticles depends on several variables, including nanoparticle concentration, size of particles, intermolecular interaction, aspect ratio, Brownian motion, clustering, purity, and interfacial thermal resistance [168]. In general, including nanoparticles can produce impressive heat transfer but will lose latent heat capacity when combined with a PCM. A variety of nanomaterials, including stable metals (Au), ceramic oxides (Al₂O₃), metallic oxides (zirconia, silica, and alumina), metallic carbides (SiC), metallic nitrides (SiN, AlN), and different carbon morphologies are commonly used for fabrication of nanocomposites [169]. Some of the widely used nanoparticles with PCMs are summarized along with their thermal conductivity, densities, and 3-D structure in Table 6. The heat transfer rates of PCM composites can be significantly improved by using metals, metal oxides, and carbon-based nanomaterials. The performance of the Nano-enhanced PCM composite was based on several parameters, which are thermal conductivity, encapsulation efficiency, TES capability, and encapsulation ratio [170].

Recent development in nano-enhanced PCMs composite is summarized in the sections below.

4.1.1. Metal-based nanoparticles

Metals are excellent conductors of both heat and electricity. Silver is a metal with the best thermal conductivity of 419 W/m-K [171]. Similarly, gold and Copper also have good thermal conductivity, but both are expensive and prone to oxidation. However, extensive research into these metals as options for enhancing PCM thermal conductivity can be seen in the literature. In a recent article, Yi et al. [172] used a silver nanowire having a diameter of 200 nm synthesized with stearic acid, as illustrated in Fig. 20(a), to form a composite with stearic acid. The TES capacity of the composite during melting was 196.4 J/g, and the heat transfer rate of composites improved due to an improvement in thermal

conductivity. The DSC results of different percentages of the silver nanowire are shown in Fig. 20(b).

Similarly, the silver nanowire was fabricated by Lv et al. [173] to form a composite with reduced graphene oxide aerogel. The investigated report shows a significant improvement in the thermal conductivity from 0.435 W/m-K (composite of lauric acid and graphene oxide aerogel) to 0.856 W/m-K (Composite of lauric acid and graphene oxide aerogel and silver nanowire) with slightly reduce in latent heat capacity. Zhang et al. [174] has synthesized paraffin with silver nanowire fabric aerogel, which has ultrahigh porosity, the composite with 5.01 wt%-Ag-NW showed an improvement in thermal conductivity was 0.35 W/m-K from 0.21 W/m-K. Furthermore, Deng et al. [175] synthesized PEG with silver nanowires, and the experiment result showed an improvement in thermal conductivity 11.3 times concerning pure PEG. A couple of research literature showed some development in copper nanowires also, as Guo et al. [176] used copper nanowires for enhancement in the thermal conductivity of composite, the experimental results show that the thermal conductivity was 20.5 times concerning paraffin. The composite PCM has also established an excellent anti-leakage property. In the article by Wen et al. [82], Cu nanoparticle of size 80–100 nm has been used to enhance the thermal conductivity of PCM. The composite PCM was made of lauric acid, nano-Cu, and attapulgite. The melting and solidification temperature results were 42.1 °C and 40.6 °C, respectively, and the latent heat of fusion was 87.1 J/g. In similar work, Sreethawong et al. [177] used hydrated CaCl₂·6H₂O salt PCM with a small number of copper nanostructures (0.17 wt%), and the experimental result showed a 52 % improvement in the thermal conductivity of the composite.

Fe metals-based filler has been used by Sun et al. [178], eicosane as a PCM, with Fe-doped carbon aerogel as a supporting matrix. The experimental result showed TES densities of 212 J/g at an onset temperature of 35 °C.

4.1.2. Metal oxide nanoparticles

Metal oxides, including alumina and copper, have good thermal conductivities ranging from 30 W/m-K to 40 W/m-K. Metal oxides have lower thermal conductivity values than metals but are also more affordable, chemically stable, and function consistently. Therefore,

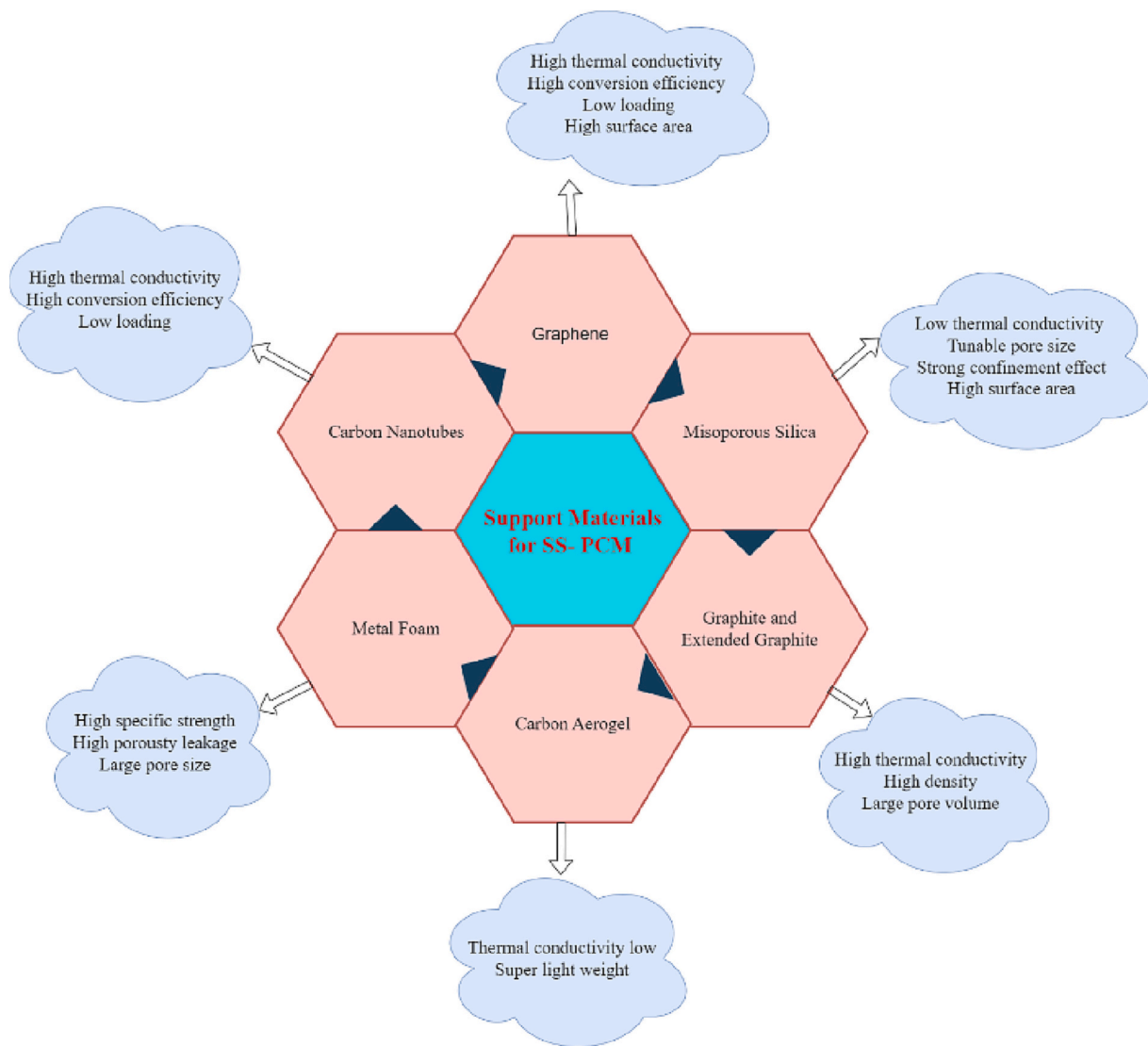


Fig. 17. The brief description of support materials for SS-PCM [164].

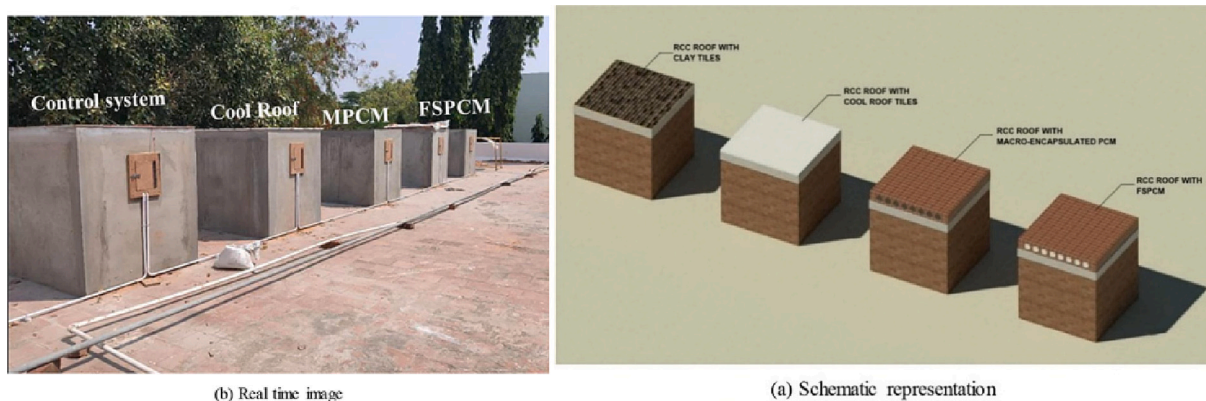


Fig. 18. Representation of prototypes used [165].

metal oxides are frequently regarded as a substitute for pure metals as additions that promote thermal conductivity. Ansu et al. [179] investigated a composite with PEG and Al_2O_3 . The experimental result showed that the thermal conductivity of PEG2000 and Al_2O_3 composite was 0.25 W/m-K with respect to 0.18 W/m-K of pure PEG2000, and the

latent heat of fusion was 157.91 J/g. Similarly, Ouikhalfan et al. [180] used Al_2O_3 , CuO, TiO_2 , and ZnO, four different metal oxides with polyethylene glycol. The experimental result showed that for two wt% of metal oxide nanoparticles loading, the thermal conductivity was 1.27–1.34 time to PEG. In a similar pattern, Khan et al. [181] used SiO_2 ,

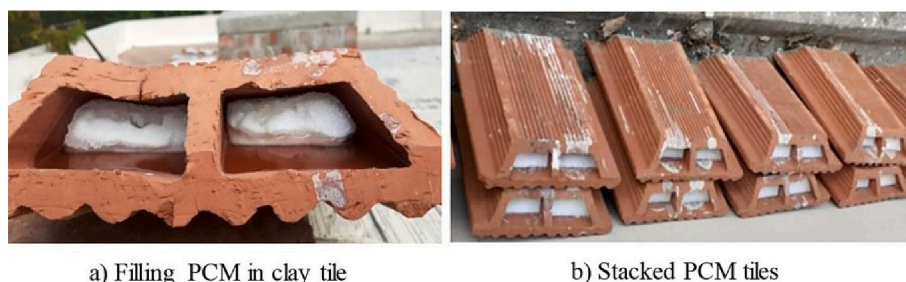


Fig. 19. PCM filled tiles for prototype building [165].

Al_2O_3 , MgO , and TiO_2 to form a composite with paraffin. The experimental result showed that the incorporation of nanoparticles increases the surface area for heat transfer and effectively improves thermal conductivity, resulting in faster charging and discharging rates. ZnO powder in paraffin as a PCM composite was used by Şahan et al. [182] to improve heat transfer performance. The shape-stabilized composite PCM was prepared in different compositions of Al_2O_3 by Mert et al. [183] and showed the TES capacity of the composite was around 86.00 J/g.

4.1.3. Carbon-based nanomaterials

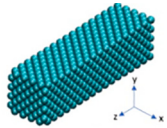
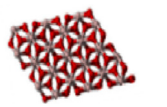
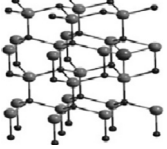
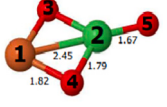
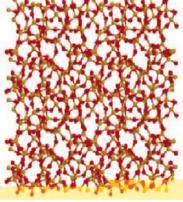
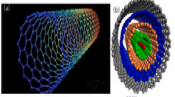
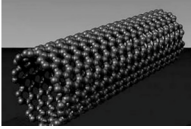
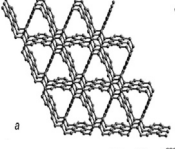
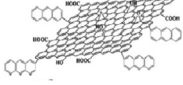
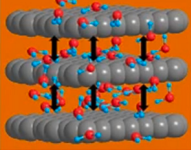
There are four categories of carbon-based materials: zero-dimensional (0D) materials, like fullerenes; one-dimensional (1D) materials, like carbon nanotubes and fibres; two-dimensional (2D) materials, like graphite and graphene; and three-dimensional (3D) materials, like carbon foam and expanded graphite [184]. Carbon-based nanomaterials have the highest thermal conductivity, nearly five times greater than metals and fifty times more than metal oxides. Carbon-derived materials are also favoured because they have the added benefit of low density [185]. The recent advancement in the field of nanotechnology is fabrication of carbon-based nanomaterials given a lead to the research community to an affordable choice for enhancing thermal conductivity. The different types of carbon-based nanomaterials commonly used as thermal conductivity enhancer are summarized with the recent development in subsequent heading.

4.1.3.1. Carbon nanotubes. Carbon nanotubes (CNTs) have cylindrical structures of carbon nanotubes (CNTs) and are constructed from layers of twisted graphite sheets. CNTs are split into single-wall carbon nanotubes (SWCNTs) and multi-wall carbon nanotubes based on the layer number (MWCNTs). In the recent development, Cheng et al. [186] used halloysite nanotube (HNT), which is a kind of naturally occurring 1:1 clay composed of outer siloxane (Si-O-Si) groups and inner aluminol (Al-OH) groups. The halloysite nanotube was further modified with 9,10-dihydro-9-oxa-10-phosphaphenanthrene-10-oxide (DOPO) to form a composite with microencapsulated PCM (Capric acid) to form D-HNT, the schematic diagram of experiment along with the synthesis of D-HNT are shown in Fig. 21(a) & (b), the test result of differential scanning calorimetry (DSC) showed that the latent heat of the composite was 113.52 J/g. Mu et al. [83] used multi-wall carbon nanotube (MWCNT) in different compositions and formed a composite with paraffin wax and high-density polyethylene. The thermal conductivity of the composite was increased with increasing the composition of the multi-wall carbon nanotube (MWCNT). The latent heat of composite material was quoted at 53 J/g at 0.5 wt% MWCNT. Mohaisen et al. [187] used PEG along with scoria and expanded perlite in different ratios. As per the result of differential scanning calorimetry, expanded perlite composite showed the highest latent heat of melting of 150.7 J/g. The expanded perlite composite was further incorporated with CNT in 0.5 wt% to improve the thermal conductivity of the composite up to 0.453 W/m-K. Two different PCMs (paraffin and $\text{CaCl}_2 \cdot 6\text{H}_2\text{O}$) and nanocarbon aerogels were synthesized as beads [188]. The result showed that by inclusion of CNT the thermal conductivity increases from 0.05 W/m-K to 0.093 W/

m-K and the latent heat of melting was 188 J/g. Harmen et al. [189] synthesized a composite of paraffin, low-density polyethylene (LDPE), and MWCNT in a ratio of 70-29-1 wt%, the result showed that the latent heat of the composite was 93 J/g and the thermal conductivity of paraffin-LDPE is increased by 28 %. Similarly, CNT was used by Cheng et al. [190] to form a composite of n-octadecane, ethylene glycol dimethacrylate alkylphenol ethoxylate, and methyl methacrylate to prepare two different PCM named C-PCM and Ref-PCM, the inclusion of CNT showed an increment in heat transfer rate. The DSC result showed the latent heat of melting of the composite was 112.25 J/g and 116.18 J/g, respectively, and 30 % of C-PCM integrated into the wall of the room was tested, and the peak temperature of the room was reduced by 11.5 °C. Liu et al. [191], used paraffin and CNT (2 wt%) with TNHDIS dispersant. The experimental result showed that the thermal conductivity increased by 18.9 % and 17.1 % compared to the paraffin before and after the thermal cycles. In a similar context, Qu et al. [192] used paraffin with hybrid Carbon Nano-additives (CNs) fillers as expanded graphite-multi-walled carbon nanotube (EG-MWCNT) and expanded graphite-carbon nanofiber (EG-CNF), the thermal conductivity of composite increased by 60 %.

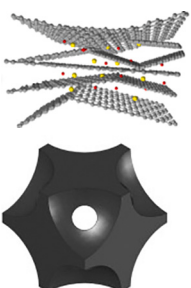
4.1.3.2. Graphene oxide. Graphene is composed of carbon atoms bound together in a hexagonal configuration. It has been experimentally studied for over 40 years. It has a significant theoretical specific surface area (2630 m^2/g), high Young's modulus (~1.0 T Pascals), and thermal conductivity (~5000 W/m-K) [193]. Graphene oxide is an amorphous carbon-based substance with a two-dimensional (2D) structure that includes hydrogen atoms and oxo functions [194]. In a recent development, a hydrothermal technique was used to create a 3D continuous network of graphene aerogel (GA) from graphene oxide (GO) [195]. A composite of PEG and GA, a ratio of 96:4 wt%, was synthesized. The experimental result showed 223.2 J/g of latent heat of melting along with 116 % enhancement in heat transfer rate due to improvement in the thermal conductivity of the composite. Pássaro et al. [196] used paraffin-wax A53 and graphene-based nanoparticles to enhance the thermal conductivity of the composite in a different ratio ranging from 0.5 to 6 wt%. For different particle sizes, the experimental result showed that the particle size of 15 μm with 6 wt% has the highest thermal conductivity enhancement of 350 %. Still, at the same time, the latent heat of fusion was reduced by 12 %. In a similar way Shamsaei et al. [197], synthesized a composite of lauric acid with graphene oxides nanosheet which sandwiches between two-dimensional (2D) calcium silicate hydrate (CSH) nanoplates, the investigated results of nanocomposites exhibited latent heat of melting was 127.6 J/g. Wu et al. [198] synthesized a shape stabilized PCM with paraffin wax (PW), melamine foam (MF), and reduced graphene oxide (RGO) MF skeletons worked as a light absorption medium as SEM and OM images of MF, MF/PW, and MF/RGO/PW as shown in Fig. 22, as the finding PW and RGO sheets can be seen to be present in the pores of MF. The composite shows a high latent heat capacity (144.8 J/g), strong solar-to-thermal energy conversion, and good thermal conductivity 0.096 W/m-K, which is 54.8 % compare with MF/PW composite. Liu et al. [199] synthesized reduced graphene oxide with three different PCMs (Stearic acid, paraffin, and

Table 6
Summary of commonly used nanomaterials in composite PCM.

Composite	Nanomaterials	Thermal conductivity (W/m K)	Density (kg/m ³)	3-D structures	Ref.
Metal nanomaterials	Silver nanowires	429	10890		[221] [222]
	Copper nanowires	2971	8960		[223] [224]
Metal oxide nanomaterials	Alumina (Al ₂ O ₃)	40	3970		[225]
	Zinc oxide (ZnO)	23.4	5606		[225]
	Ferric oxide (Fe ₂ O ₃)	15	5240		[226]
	Silica (SiO ₂)	1.296	2650		[227] [228]
Carbon-based nanomaterials	Carbon nanotubes	3000	2600		[225]
	Carbon nanofibers	4000	2260		[225] [200]
	Graphite	70	799		[229] [230]
	Graphene oxide	3000	3600		[225]
	Exfoliated graphite nanoplatelets	2–300	5.3–10		[231] [232]

(continued on next page)

Table 6 (continued)

Composite	Nanomaterials	Thermal conductivity (W/m K)	Density (kg/m ³)	3-D structures	Ref.
	Expanded graphite	4–100	300		[233] [234]
	Carbon foam	43–150	500		[235]

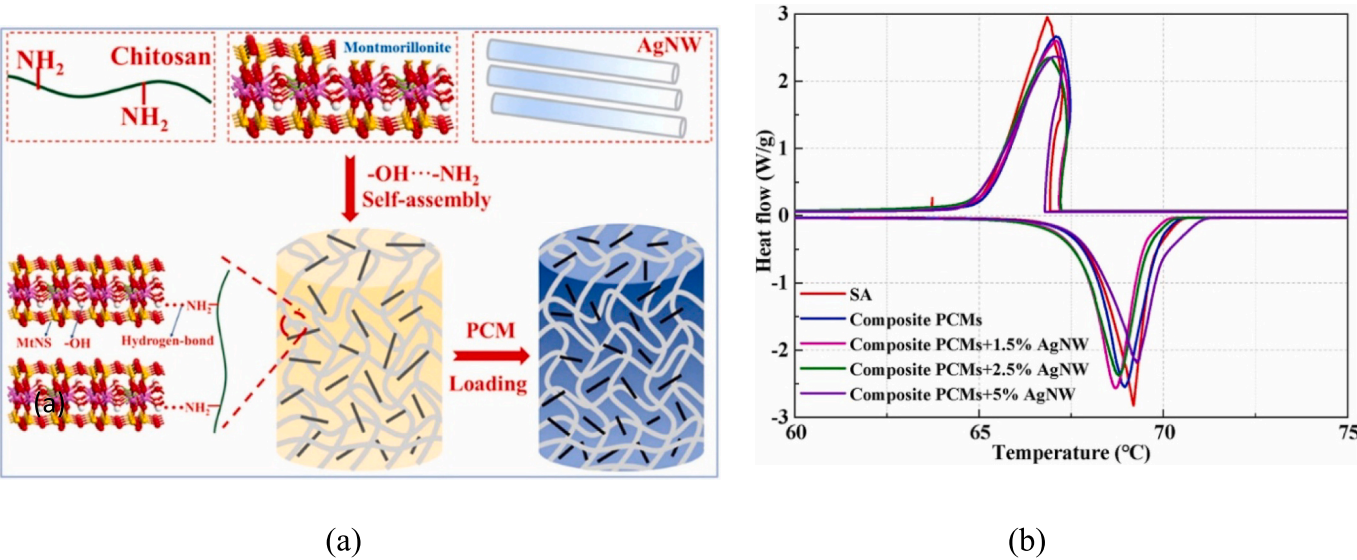


Fig. 20. DSC results of different percentages of silver nanowires [172].

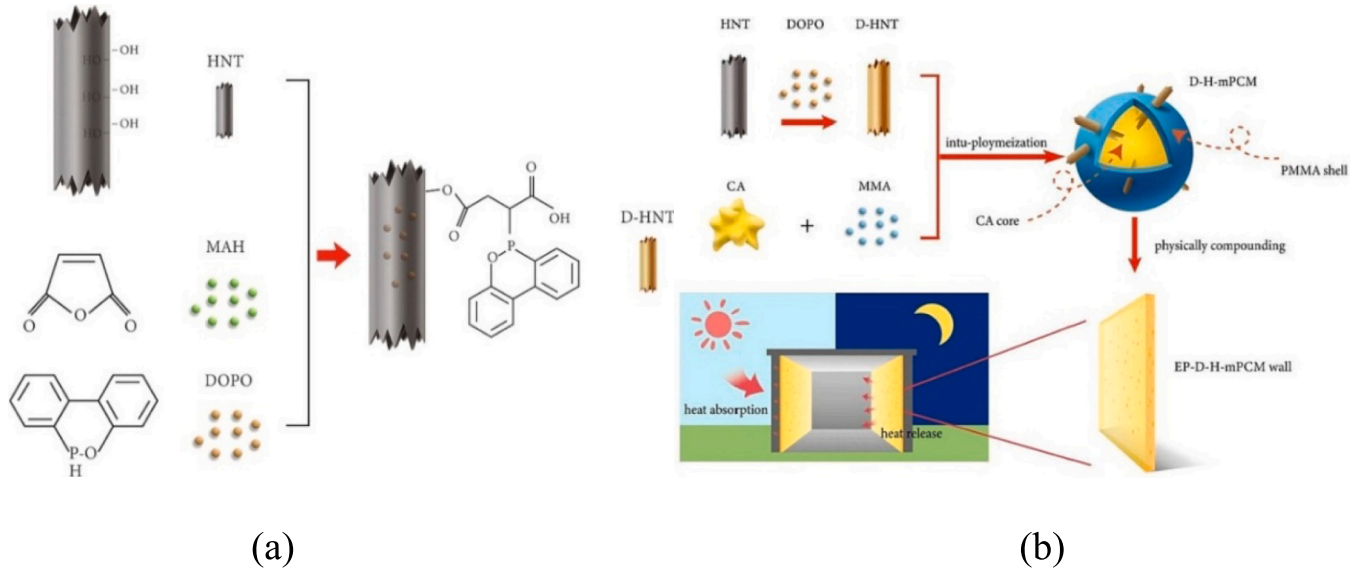


Fig. 21. (a) Synthesis D-HNT, (b) schematic diagram of experiment [186].

octadecanol) in the experiment result, the stearic acid with reduced graphene oxide showed good latent heat of 142.8 J/g with a thermal conductivity of 3.21 W/m-K, and further thermal conductivity can enhance up to 5.89 by including Ag.

4.1.3.3. Carbon fibres. Carbon fibre is an inorganic fibre with a carbon

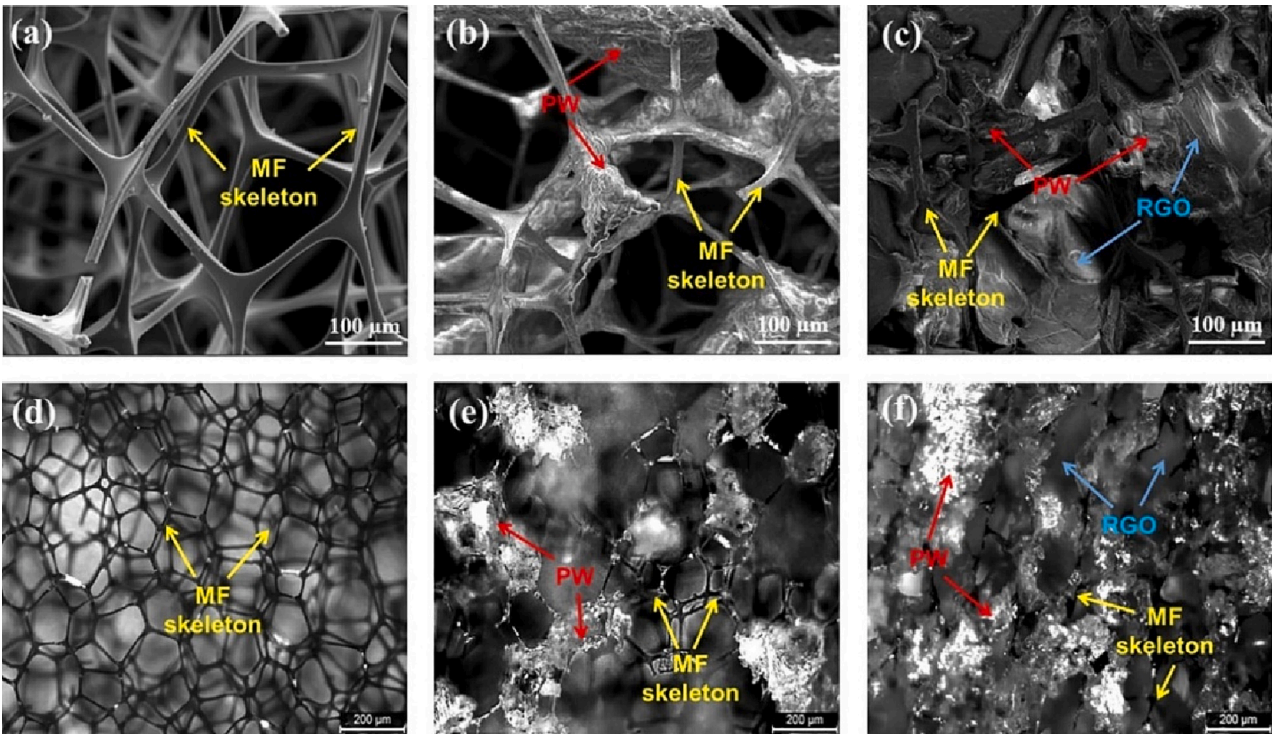


Fig. 22. SEM and OM image of MF, MF/PW, and MF/RGO/PW [198].

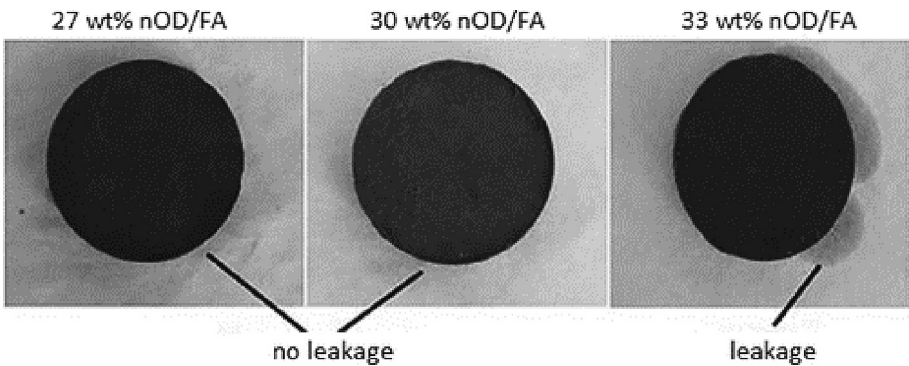


Fig. 23. The different composition of PCM composite of n-octadecane and fly ash specimen for leakage test [203].

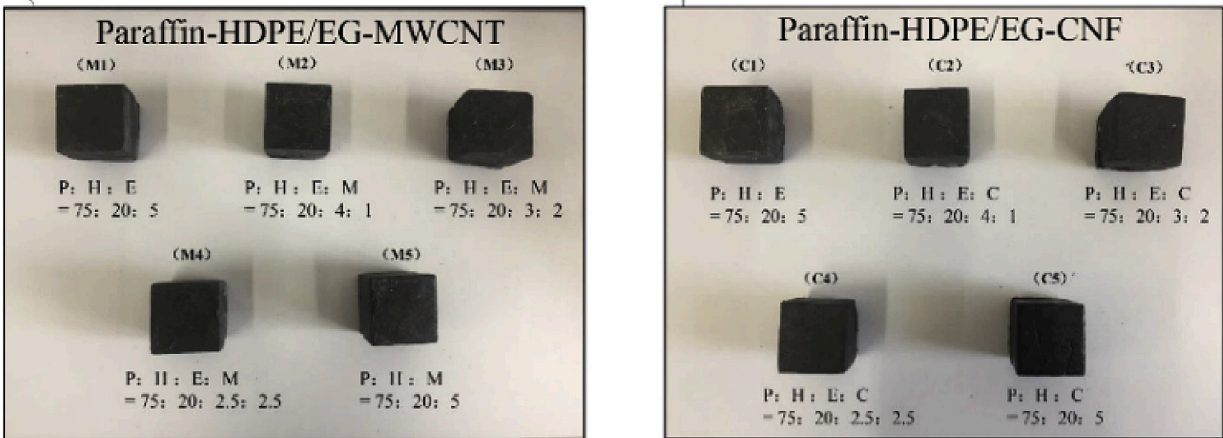


Fig. 24. Paraffin -HDPE/EG-MWCNT and CNF with different compositions [204].

Table 7

Summary of the recent development in TES systems by using PCM with nanofiller.

PCM	Nanofiller	% Weight of nanofiller	Melting temperature (°C)	Thermal conductivity (W/m-K)	Latent heat of melting (J/g)	Ref.
PureTemp 23	Graphene powder	3	22.7	–	111.7	[208]
Lauric acid	Expanded graphite	1.5	19.5	0.393	144.52	[209]
n-Octadecane	Expanded graphite	25	24.1	20.6	148	[210]
Stearic acid	Boron nitride and expanded graphite	12	67.85	6.349	154.20	[211]
Paraffin	Silver nanowire fabric aerogel	5.01	–	0.35	–	[174]
Dodecanol	Carbon nanofiber	10	18.04	1.01	157.03	[201]
Disodium hydrogen phosphate dodecahydrate	CNF-EG	1.27-1.5 CNF & 13.8 EG	33.7	2.486	208.9	[206]
PEG2000	Al ₂ O ₃	–	–	0.25	157.91	[179]
Polyethylene glycol	Expanded perlite with CNT	0.5	–	0.453	150.7	[187]
N-octadecane	CNT	–	28.97	–	112.25	[190]
N-octadecane and fly ash	CNT	8	26.03	0.79	62.71	[236]
N-octadecane and fly ash	CNF graphene nanoplatelet	8	26	0.63	60.5	[236]
N-octadecane and fly ash	Graphene nanoplatelet	8	26.43	0.84	62.71	[236]
Octadecanol	Reduced graphene oxide	10	–	2.21	165.8	[199]
Stearic acid	Reduced graphene oxide	10	–	3.21	165.3	[199]
Polyethylene glycol	Graphene oxide	4	64.0	–	142.8	[237]
Sodium acetate trihydrate	Expanded graphite	5	37.6	0.31	101	[212]
Stearic acid	Graphitic hierarchical porous carbon	15	67.5 °C	0.879	171.5	[238]
Paraffin	Expanded graphite	6	26.49	2.141	142.39	[213]
Palm kernel vegetable fat	Graphite	–	27.33	0.86	40.27	[214]
Paraffin	Expanded graphite and CNF	9.4	22.3	5	105.3	[239]

content of greater than 90 % by weight. A significant aspect ratio of carbon fibres (CNF) can be produced when the SP² bond in the graphite grows along the axial direction parallel to the base surface. CNF have diameters ranging from 2 to 100 nm and can be up to 100 µm long [200]. Synthesis of CNF is comparatively easy and cost-effective with CNT, but it has low thermal conductivity. In recent development, Hekimoğlu et al. [201] synthesized polyacrylic acid (PAA), dodecanol (DDA), and carbon nanofiber (CNF) by 10 wt%. The experimental result showed the melting at 18.04 °C and having a phase change enthalpy of 157.03 J/g, the thermal conductivity of the composite enhanced by 179.77 %. Hekimoğlu et al. [202], a composite of silica fume and PEG were investigated, the melting temperature was around 14–15 °C, and its latent heat of fusion was about 55–56 J/g. Furthermore, the filler of different nanoparticles, carbon nanofiber (CNF), carbon nanotube (CNT), and graphene nanoplatelet (GNP), were incorporated to enhance the thermal conductivity of composite with nanofiber as filler of amount 3 part per million of fibre showed a thermal conductivity of 0.351 W/m-K. Hekimoğlu et al. [203] performs another study where a shape-stabilized PCM composite of n-octadecane and fly ash was prepared. Leakage test showed that the maximum 30 wt% of PCM could be held in fly ash without leakage as shown in Fig. 23. The latent heat of fusion and the composite's thermal conductivity were 64.10 J/g and 0.31 W/m-K, respectively. Furthermore, the Nano based filler was introduced to enhance the thermal conductivity of the composite, CNF, CNT, and graphene were used.

Qu et al. [204] synthesized a composite of paraffin and high-density polyethylene (HDPE) and experimented with the hybrid composition of expanded graphite (EG) with CNF and EG with MWCNT filler in different ratios, as shown in Fig. 24. The thermal conductivity of the M2 (75 % paraffin, 20 % HDPE, 4 % EG, and 1 % MWCNT) and C2 (75 % paraffin, 20 % HDPE, 4 % EG, and 1 % CNF) samples was highest at 1.36 and 1.03 W/m-K respectively, and a similar experiment was also performed by [205]. Hu et al. [206] developed inorganic hydrated salt (disodium hydrogen phosphate dodecahydrate) filled with CNF and EG in hybrid form to enhance the thermal conductivity. The latent heat of fusion and the composite's thermal conductivity were 208.9 J/g and 2.486 W/m-K 1.27 wt% and 1.5 wt% of CNF respectively.

4.1.3.4. Graphite and expanded graphite. Graphite has a layered

structure of sheets of horizontally spaced rings of six carbon atoms, which have a metallic dark grey colour. It has good thermal conductivity (100-400 W/m-K) and is less expensive than other carbon additives [207]. A modified kind of graphite called expanded graphite has layers and interlayer spaces. It has a thermal conductivity of 4–100 W/m-K, in the recent development of graphite powder used as a thermal conductivity enhancer by [208], PureTemp 23, a commercial grade PCM. The experimental result showed that the latent heat of melting was 111.7 J/g, and the heat transfer rate increased by 50 %. The expanded graphite and lauric acid composite were integrated with tiles by Wu et al. [209], the experiment result showed that the thermal conductivity of the composite in solid and liquid phases was 0.690 & 0.393 W/m-K respectively, and the latent heat of melting was 144.52 J/g. An et al. [84] investigated thermal properties of a composite of octadecanoic acid, octadecanol, and expanded graphite. Expanded graphite increases the thermal conductivity from 0.066 to 3.38 W/m-K, and the latent heat of storage has been 135-149 J/kg. Wu et al. [210] prepared a composite of n-Octadecane olefin block copolymer as supporting material and expanded graphite in the different ratios experimented with the conductivity and the latent heat of composite 20.6 W/m-K and 148 J/g, respectively. Ao et al. [211] used stearic acid with expanded graphite and boron nitride (BN) to synthesize a composite for TES application. The thermal conductivity of the composite was 6.349 W/m-K when the mass fraction of BN and EG was 12 wt%. Yang et al. [212] used sodium acetate trihydrate as PCM with a different type of carbon additive, added ammonium chloride as a phase change temperature regulator, and the degree of supercooling decreased by 0.9 °C. Ren et al. [213] used epoxy resin-sealed expanded graphite/paraffin composite. The experimental result showed that the LHS was 142.39 J/g at a melting temperature of 26.49 °C, and the thermal conductivity of the composite was 2.141 W/m-K. The hydrogenated Palm Kernel vegetable fat PCM was used by Boussaba et al. [214]. The latent heat capacity was 40.27 J/g at a melting temperature of 27.33 °C to enhance the thermal conductivity of PCM graphite in the different ratio was used, and the maximum thermal conductivity was 0.86 W/m-K. Guo et al. [215] investigated a composite of wood flour, high-density polyethylene, and paraffin PCM composite with expanded graphite by 2 wt%, the experimental result showed the latent heat capacity of 26.8 J/g at a peak temperature of 23.3 °C and the thermal conductivity enhanced by 43 % at 20 °C. Wen et al. [216]

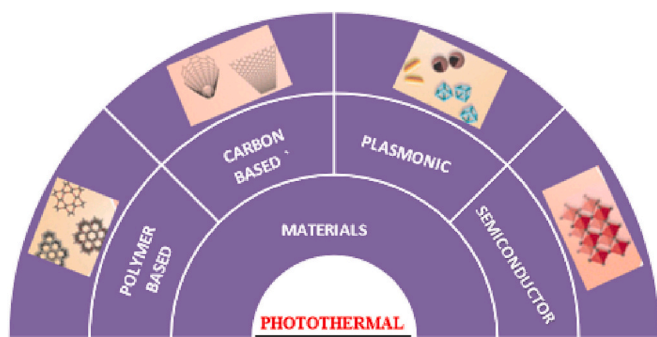


Fig. 25. Classification of photothermal materials [242].

prepared a eutectic mixture of capric-lauric acid supported by diatomite which shows latent heat capacity of 87.33 J/g with a melting temperature of 23.61 °C. The composite shows excellent thermal durability and thermal reliability at elevated temperatures. Further, to enhance the thermal conductivity of the composite, expanded graphite was added in different compositions of 3 %, 5 %, 7 %, and 10 % which results in increase in thermal conductivity of the composites by 39.7 %, 61.6 %, 77.6 %, and 114.2 % respectively. Liu et al. [217] used paraffin with expanded graphite to form a cement block to enhance the TES capacity. The latent heat of the cement block composite was 34.87 J/g. Similarly, Lin et al. [218] used carbon foam as a thermal conductivity enhancer with PEG as PCM. The experimental result shows a LHS capacity of 135.7 J/g, and the thermal conductivity improved by 109.8 %. Yu et al. [86] prepared a novel TES cement mortar using eutectic hydrated salt coated with PEG and perforated FAC (Fly ash cenosphere) by vacuum adsorption, and the composite showed high latent heat of fusion (129.7 J/g) and melting temperature of 15.2 and 23.5 °C. In recent work, Raju et al. [80] used n-octadecane/melamine formaldehyde nanocapsules of size range 76 nm to 530 nm embedded with cement mortar. The result

shows that the melting temperature and latent heat of fusion were 26.01 °C and 122.24 kJ/kg, respectively, but the compressive strength was compromised by increasing 6 wt% of PCM and reducing by 44.41 %. Neopentyl glycol as a supporting media for Docosane was evaluated by [219]. Expanded graphite was also loaded in the composite to improve heat transfer. The results show maximum loading of 60 wt% of Docosane and the composite obtained excellent latent heat capacity of 210.74 J/g. A study by Sani et al. [220] uses Crodatherm as a water-insoluble organic PCM derived from plant sources and coated with graphite powder. The latent heat of storage for PCM was 174 J/g at a melting temperature of 10.24 °C. Table 7 summarized the recent development in TES systems by using PCM with different nanofiller.

4.2. Solar to thermal performance of PCM

Photothermal materials are nanomaterials with well-defined dimensions, shapes, compositions, and surface functions that allow low-density light energy to be converted to thermal energy. The fundamental working principle of photothermal conversion is that photothermal fillers can be activated by light, causing electrons to jump to the excited state. Then, nonradiative relaxation, energy transfer, or quenching will allow the excited electrons to release their energy and return to the ground state [240]. The most critical considerations for photothermal applications are material and structural design. Several Near Infra-Red (NIR) responsive photothermal materials and structural engineering techniques have recently been created to suit various applications [241]. Photothermal materials can be classified into four basic categories based on their functionality, semiconductors, plasmonic metals, carbon, and polymer-based materials, as shown in Fig. 25 [242]. The three basic photothermal mechanisms for conversions are electron-hole generation and relaxation of semiconductors, as shown in Fig. 26 (b), plasmonic localized heating, as shown in Fig. 26(a), and HOMO-LUMO excitation and lattice vibration of molecules, shown in Fig. 26 (c). The highest energy molecular orbital with electrons is called the

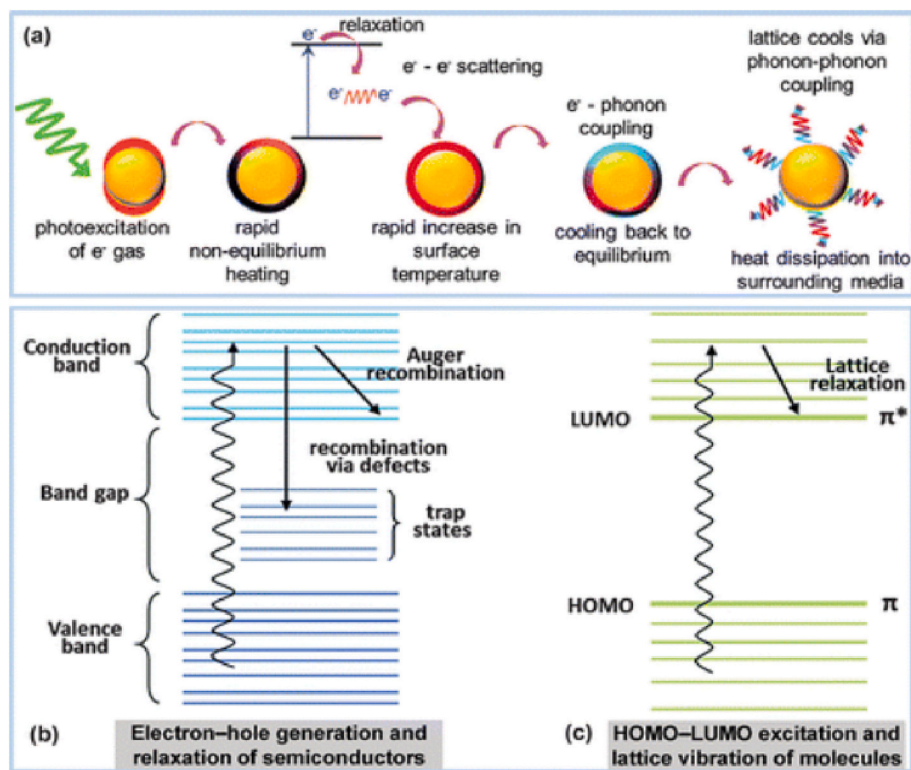


Fig. 26. (a) Plasmonic localized heating (b) electron-hole generation and relaxation of semiconductors (c) and HOMO-LUMO excitation and lattice vibration of molecules [242].

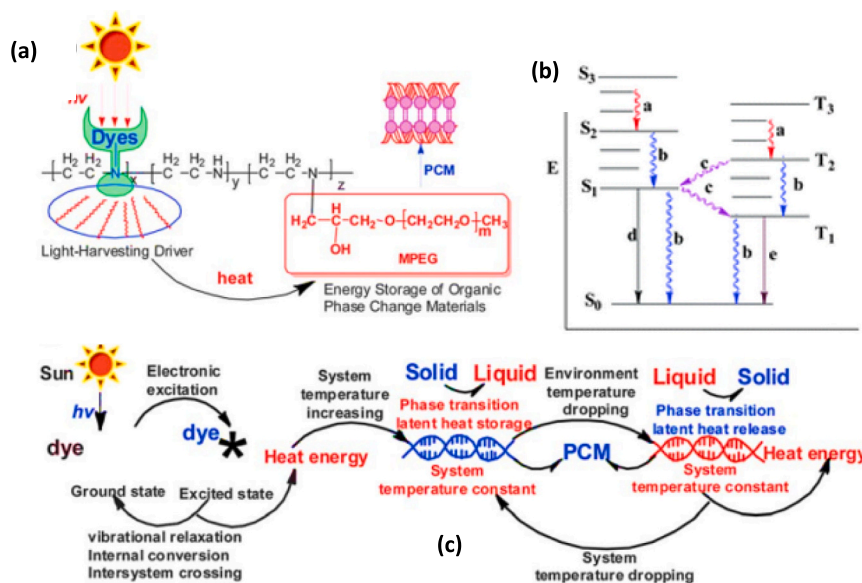


Fig. 27. (a) Dye as an energy absorber and converter (b) dye molecules excitation and relaxation (c) solar to thermal conversion through dye [251].

HOMO, while the next highest energy molecular orbital with no electrons is called the LUMO. The excitation and lattice vibration between these two molecular orbitals work as the photothermal conversion. Photothermal materials have a high rate of energy conversion efficiency and require less preparation. According to their constituent parts, they can also be classified as carbon-based materials, metal-based materials, organic materials, and other photothermal materials [243]. Due to their capacity to absorb a broad spectrum of sunlight, carbon-based materials are a significant focus in the recent study of photothermal materials. It strongly absorbs portions of visible and near-infrared light and simultaneously benefits from excellent stability and low cost [244]. Another class of photothermal material that exhibits the local surface plasma resonance (LSPR) phenomenon that may absorb longer-wavelength infrared light and transform it into thermal energy is metal-based materials [245]. If the frequency of the incident light and the resonance frequency of the free electrons on the surface of the metal particle match, coherent oscillations will cause the collective excitation of the electrons to occur when a light ray strikes a metal-based nanoparticle. Organic photothermal conversion materials have high photo absorption, photostability, and photothermal conversion efficiency. Compared to carbon-based materials, they are simpler to create and process and their ability to control light absorption may be altered across a large range [246]. Some newly developed photothermal materials, such MXene, due to the LSPR effect, display remarkable photothermal performance [247].

The performance of photothermal materials is measured by thermal flux and photothermal efficiency. The photothermal conversion efficiency is determined by the fraction of the absorption in the extinction [248]. The thermal flux and photothermal efficiency can be calculated by using Eqs. (6) and (7); [249].

$$\Phi = \frac{m \Delta H}{t_e - t_i} \quad (6)$$

$$\eta = \frac{m \Delta H}{E_d S (t_e - t_i)} \quad (7)$$

The different parameters in the equations are as follows, m is the sample mass, ΔH is the enthalpy of phase transition, t_e , and t_i the phase transition temperature at the end and the beginning, respectively, E_d is the energy density, S exposure area under illumination.

There are different areas of applications of photothermal materials, like photothermal therapy, photothermal sterilization, and solar evaporation. In recent years the energy conversion and storage based on PCM

with the photothermal effect has received much attention. The ability of PCMs to effectively absorb incident sunlight can be improved by including photothermal elements within them. Additionally, it should be noted that based on their chemical composition, each photothermal material can only absorb a particular range of light wavelengths. The photo absorber and converter are crucial components of photothermal conversion. PCMs and dyes have great promise as photothermal materials, as dyes have great potential as photo converters and absorbers [250]. The dyes and PCMs are composite reactive agents that serve as light-capture and TES components, as shown in Fig. 27(a) & (c) [251]. By absorbing photons, the dye molecules become activated. Then, using the thermal release strategy, they quickly shift to the ground state and relax their vibrations see Fig. 27(b), and the heat energy is released. Zhao et al. [252] has examined a composite material, carbon-coated aluminosilicate nanolayers created by intercalating kaolinite with potassium acetate and subsequent calcination. The photothermal conversion efficiency of the composite was coated as 92.1 %. Yang et al. [253] conducted an experiment where composite PCMs with zirconium carbide contents showed excellent heat transport properties, photo absorption, and high thermal storage capacity. When doped with 0.01 wt% zirconium carbide, the photothermal conversion efficiency can reach up to 81 %. In a similar context, a composite of Carbon nanotube/nickel foam-mannitol prepared by Wang et al. [254]. Under the optical power intensity of 700 m-W/m², the photothermal conversion efficiency was 86 % and showed a heat storage capacity of 314.4 J/g. Sun et al. [178] investigated eicosane as a PCM with Fe-doped carbon aerogel as a supporting matrix. The experimental result showed that under the optical power intensity of 200 m-W/m², 250 m-W/m², and 300 m-W/m², the thermal conversion efficiency was 60.63 %, 77 %, and 93.32 %, respectively, and the TES densities of 212 J/g at onset temperature of 35 °C. Zhang et al. [255] used PEG supported on silver particles and graphene nanosheets to improve solar to thermal performance. The PCM's photothermal efficiency was 88.7 % and it had a 166.1 J/g energy storage density. The thermal conductivity of the composite also improved by 49.5 % about the PEG6000. A similar composite of PEG with Fe₃O₄ - GNS with the carbonized wood floor was prepared by Chao et al. [249] and improved the photothermal and thermal conductivity. Kong et al. [256] used industrial paraffin wax with polypyrrole wrapped over a Polydivinyl benzene nanotube. It shows a photothermal efficiency of 85.2 % and an energy storage capacity of 145.7 J/g.

Wang et al. [257] prepared a composite of PEG with carboxylate

Table 8

Summary of the recent development in photothermal materials.

PCM	Filler	Filler (% wt.)	Conversion efficiency	Thermal conductivity (w/m K)	Enthalpy of melting (J/g)	Optical power intensity (m-W/m ²)	Ref.
Mannitol	CNT	15	86 %	2.19	314.4	700	[254]
Mannitol	CNT	10	81	–	312.9		
Mannitol	CNT	5	56.7	–	308.2		
Mannitol	–	–	34.9	0.88	316.6		
PW	–	–	49.6	0.28	147.8	250	[273]
PW	MS@RGO	–	76.7	0.54	137.2		
PW	MS@RGO ZrC	0.02	73.9	0.62	130.1		
PW	MS@RGO ZrC	0.01	81	0.58	134		
PW	MS@RGO ZrC	0.006	80	0.57	136.4		
PW	MS@RGO ZrC	0.003	78.5	0.56	135.8		
Dodecanoic acid	Graphene	30	78	0.5	137.8	500	[274]
Paraffin	EG	–	88	1.1	138.7	300	[267]
PEG	GO	5	75.9	–	87.36	200	[268]
PEG	GO	10	82.5	–	86.58		
PEG	GO	15	87.3	–	84.27		
PEG6000	–	–	–	0.212	185.3	120	[255]
PEG	GNS	2	70.8	0.257	176.9		
PEG	Ag-GNS	2	88.7	0.317	177.2		
PEG	Ag-GNS	4	89.6	0.337	173.3		
PEG	Ag-GNS	6	90.9	0.367	169.6		
PEG	Ag-GNS	8	92	0.414	166.1		
Industrial paraffin wax (IPW)	Polydivinyl benzene nanotube	–	85.2	–	145.7	100	[256]
n-Octacosane	CNTs	4	83.4	0.6	252.4	300	[263]
Paraffin	–	–	–	0.25	197.89		[275]
Paraffin @SnO ₂	–	–	75	0.69	132.17	100	
Paraffin @SnO ₂	CNTs	5	91.79	0.77	85.67	100	
PW	3-D RGO	10	85	0.46	170.4	1,850,000	[276]
PW	–	–	–	0.25	189.5		
PEG2000	–	–	0	0.252	170.4	70	[257]
PEG 2000	GO	31.4	47.9	–	96.9		
PEG 2000	Blue anthraquinone dyes (Bdye) and GO	30.9	62.9	–	109.4		
PEG 2000	Bdye-grafted GO	19.6	86.6	0.499	130.6		
Mg (NO ₃) ₂ ·6H ₂ O (MCH)	Ti ₄ O ₇	10	78.6	0.59	192.17	–	[277]
MCH	Ti ₄ O ₇	7	–	0.56	202.88		
MCH	Ti ₄ O ₇	5	–	0.54	205.01		
MCH	Ti ₄ O ₇	3	–	0.52	208.89		
MCH	Ti ₄ O ₇	1	–	0.49	215.14		
Paraffin	Graphene aerogel and copper foam	–	97	2	133.62	–	[260]
Paraffin	Graphene aerogel and copper foam	–	95.98	0.28	154.64	250	[278]
Paraffin	–	–	49.6	–	147.8	250	[270]
Paraffin	ZrC	0	76.7	–	137.2		
Paraffin	ZrC	0.07	78.5	–	135.8		
Paraffin	ZrC	0.14	80	–	136.4		
Paraffin	ZrC	0.23	81	–	134		
Paraffin	ZrC	0.86	73.9	–	130.1		
PEG	–	–	–	0.23	212.8	–	[261]
PEG	Carboxylic multiwalled carbon nanotube (CMCNT)	0.37	85.8	0.33	132.5		
PEG	CMCNT	1	86.6	0.39	129.3		
PEG	CMCNT	2	87.5	0.44	125.2		
PEG	CMCNT	3	88.3	0.48	116.3		
PEG 6000	CNTs	40	87.7	1.5	107.4	100	[279]
PEG	–	–	–	–	210.9	100	[280]
PEG	Melamine foam	–	–	0.087	189.7		
PEG	Mxene-coated melamine	–	92.7	0.146	183.5		
PEG	carbon foam	10	–	0.29	–	–	
PEG	Hydroxyl boron nitride nanosheets (H-BNN)	5	–	–	158.7		[281]
PEG	@ CuO	10	91.8	1.12	152.3		
PEG	@ CuO	15	–	–	148		
Paraffin and 70 % with thermoplastic elastomer 30 %	Expanded graphite	–	72	2.2	145.4	300	[266]
Eicosane	Fe-doped carbon aerogel	–	93.32	–	212.	350	[178]
Cellulose/poly-(hexadecyl acrylate) polymer	Graphene	5	74	0.7	103	100	[264]

(continued on next page)

Table 8 (continued)

PCM	Filler	Filler (% wt.)	Conversion efficiency	Thermal conductivity (w/m K)	Enthalpy of melting (J/g)	Optical power intensity (m-W/m ²)	Ref.
Cellulose/poly-(hexadecyl acrylate)	Graphene	7	81	0.95	90	100	
Cellulose/poly-(hexadecyl acrylate) polymer/	Graphene	9	89	1.3	78	100	
n-Octadecane	Graphene/SiO ₂	0	42	0.6416	109.7	150	[269]
n-Octadecane	Graphene/SiO ₂	0.25	–	0.7471	108.4		
n-Octadecane	Graphene/SiO ₂	0.75	–	0.7987	109.1		
n-Octadecane	Graphene/SiO ₂	1.5	70	1.4941	108.8		
n-Octadecane	Graphene/SiO ₂	3.75	–	0.9953	106.1		
n-Octadecane	Graphene/SiO ₂	5	–	0.9719	107.5		
Paraffin	GNP/GO	4.89	88	1.46	157.5	100	[271]
Paraffin	Hierarchical porous carbon foams	–	91	–	120.2	110	[282]
PEG	Hierarchical porous carbon foams	–	87	–	163.3	110	
Paraffin wax	Hierarchical porous carbon	2	31	0.3	152.2	200	[283]
Paraffin wax	Hierarchical porous carbon	2	38	0.3	152.2	250	
Paraffin wax	Hierarchical porous carbon	2	89	0.3	152.2	300	
1-Hexadecanamine	carbon aerogels	–	67.8	0.373	217.1	100	[272]
1-Hexadecanamine	carbon aerogels	–	75.6	0.379	239.4		
Paraffin	CNTs sponge	20	41.9	1.2	131.7	58	[284]
	CNTs sponge	20	54	1.2	131.7	70	
	CNTs sponge	20	60.2	1.2	131.7	90	
Palmitic acid	Carbon foam	–	65	–	124	90	[285]
Palmitic acid	Carbon foam	–	82	–	124	100	
Palmitic acid	Carbon foam	–	95	–	124	110	
Lauric acid	Graphene aerogel & RGO	5	80.6	1.2	207.3	100	[265]
Lauric acid	Graphene aerogel & RGO	5	79.5	1	208.5	100	
Lauric acid	Graphene aerogel & RGO	5	74.6	0.9	206.6	100	
PEG	Graphene aerogel	1.32	80.1	–	108.1	100	[286]
PEG 4000	Graphite foam	24	66.9	3.5	64.5	90	[287]
PEG 6000	Graphite foam	18	66.2	3.5	76.1		
PEG 8000	Graphite foam	19	43.8	3.5	80.3		
n-Eicosane	Ti ₂ O ₃ /carbon foam	2.5 vol%	72.8	–	200.1	87	[288]
n-Eicosane	Ti ₂ O ₃ /carbon foam	5 vol%	85	–			
n-Eicosane	Ti ₂ O ₃ /carbon foam	7.5 vol%	88.2	–			
PEG4000	MXene	–	92.5	–	–	100	[289]
PEG4000	MXene	–	89	–	–		
PEG4000	MXene	–	73.7	–	–		
PEG4000	MXene	–	55.6	–	–		
PEG10000	MXene	–	85.5	–	–		
PEG10000	MXene	–	87.9	–	–		
PEG6000	MXene	20	94.2	0.3	157.9	128	[290]
PEG6000	MXene	15	85.1	0.3	171.5	128	
PEG6000	Graphene nanoparticles/Fe ₃ O ₄	1	92	0.4	101.5	100	[291]
PEG6000	Graphene nanoparticles/Fe ₃ O ₄	4	78.4	0.3	109.8		

graphene oxide and blue anthraquinone dye, and the experimental result showed a photothermal efficiency of 86.6 % and a thermal conductivity of 0.499 W/m-K. Hu et al. [258] synthesize flexible textile coated with polypyrrole micro PCM. The calculated photothermal efficiency was 94.03 % under the intensity of 0.5 kW/m² with a thermal capacity of 231.6 J/g. Xiong et al. [259] prepared a shape stabilized PCM composite using PEG with graphene oxide. It was found that 4 wt% of graphene oxide will prevent the leakage of PCM. The composite shows heat storage capacity of 174.5 J/g having 95.6 % of PEG. Zheng et al. [260] conducted a study where the PCM composite of paraffin (PA) with graphene aerogel (GA) and copper foam (CF) was examined with reported photothermal efficiency of 97 % for a composition of PA60/GA30/CF. Yan et al. [261] uses PEG and SiO₂ carboxylic multi-walled carbon nanotube. The photothermal efficiencies for different compositions ranged from 85.8 to 88.3 %.

Yang et al. [262] performs a study where paraffin wax with graphene aerogel were used to form a composite PCM with a photothermal efficiency of 95.98 % at the intensity of 250 mW/m². Du et al. [263] used CNTs, cellulose hybrid aerogel with n-Octacosane composite as solar to thermal energy conversion with a conversion efficiency of 83.4 %. Qian et al. [264] synthesized a composite of graphene, cellulose, and Poly (hexadecyl acrylate) in a different composition. The experimental result

showed that the composite's solar-to-thermal energy conversion and storage efficiencies hold excellent potential for usage in solar energy collection systems and thermal energy management applications. Mu et al. [265] prepared a form stable composite PCM of lauric acid, graphene aerogel, and reduced graphene oxides. The investigated results showed that the composite exhibited high photon absorption and had a light-to-heat conversion efficiency of 80.6 %. Wang et al. [266] used a paraffin composite and expanded graphite with thermoplastic elastomer in a different ratio. The composite ratio of 70 % and 30 % showed excellent latent heat capacity and photothermal conversion efficiency. In a similar pattern, Chen et al. [267] prepared a composite of 70 % paraffin and expanded graphite with different ratios of polyethylene and ethylene-propylene rubber. The experimental result showed a latent heat capacity of 138.7 J/g and photothermal conversion efficiency of 88 %. Li et al. [268] used paraffin wax with graphene oxide as photothermal material, and the results showed a good latent heat capacity with a photothermal conversion efficiency of 87.3 %. Zhu et al. [269] investigated a slurry based on graphene nanoparticle PCM. The photothermal conversion efficiency increased from 41 % for water to 70 % for the slurry. The addition of graphene-enhanced the thermal conductivity by 132.9 %. Yang et al. [270] used zirconium carbide as a photothermal absorber with paraffin and reduced graphene oxide. The experimental

results showed that the photothermal conversion efficiency increased by increasing the wt% of zirconium carbide. Xue et al. [271] investigated a GO and GNP fill with PW. The experiment showed a photothermal thermal conversion efficiency of 88 % with a latent heat capacity of melting of 157.5 J/g. Wang et al. [272] used biomass carbon aerogels with 1-hexadecanamine showed good solar thermal conversion efficiency. Table 8 summarized the brief details of solar to thermal conversion performance of recent work.

5. Research outlook

The development in nanotechnology played a significant role in improving thermal and physical performance of PCM. Carbon-based nanomaterials are in trend for improving the thermal conductivity of PCM because of significantly high thermal conductivity, nearly five times greater than metals and almost fifty times more than metal oxides. Nanomaterials, such as carbon nanotubes (CNTs), graphene, graphite, expanded graphite, graphene nanoplatelets, and metallic nanoparticles, are incorporated into PCMs to enhance their thermal conductivity. These nano additives create a conductive network within the PCM, facilitating faster heat transfer and improving the overall performance of the thermal energy storage system. Research should focus on understanding the interaction mechanism between nanomaterials, surfactants, and PCM in nanomaterials-based composite PCM. By studying these interactions, optimizing the properties of nanomaterials-based composite PCM and improving their thermal performance will be possible.

The current trend of improving shape stability of the PCM involves incorporating PCM within a matrix or host material to enhance mechanical strength, stability, and heat transfer properties. These methods include embedding PCM in polymers, foams, or porous materials to create composites with improved thermal characteristics and mechanical integrity. Further investigation of the porous media and PCM interaction in composite PCM systems is needed. The geometric parameters of the porous media, such as pore density and porosity, significantly influence the thermal properties of composite PCM. Optimizing the porous media configuration will enhance the thermal properties of TES systems. Despite the considerable efficiency of photothermal materials, some challenges such as finding low-cost, nontoxic, environmentally friendly, readily available, and recyclable materials must be addressed. Additionally, the intermittent nature of solar radiation, weather changes, and scaling of solar absorbing materials need to be considered to optimize photothermal materials performance in TES systems.

6. Conclusion

TES using PCM can help in mitigating the challenges of poor performance of solar energy technologies. High energy density and minimum temperature change makes them one of the potential TES material. PCM stores the excess amount of thermal energy available during the sunny days and releases it as and when required. However, there are few major thermo-physical limitations of PCM which limits the applicability. Literature review shows that almost all the PCM shows poor solar to thermal conversion performance, low thermal conductivity, leakage during phase transformation and high degree of super cooling. Numerous attempts have been made to overcome these drawbacks to improve thermo-physical performance of PCM for improved TES. Leakage during phase transition leads to loss of the PCM from the TES system. To improve stability and minimize leakage, microencapsulation and macroencapsulation techniques are commonly used. Encapsulation of PCM has significantly reduces the leakage of the PCM and also improves heat transfer performance because of increased in surface area. Various porous materials were also used to stabilize the shape of the PCM during phase transformation. Poor thermal conductivity of the PCM causes slow charging and discharging. Development of composite PCM loaded with metal nanoparticles, metal oxide nanoparticles, and carbon

based nanoparticles have given a significant boost in improving thermal conductivity. Solar to thermal performance of the PCM can be improved by loading photothermal fillers. Photothermal materials have high solar to thermal conversion efficiency. Thus, when these photothermal fillers loaded in the PCM they form composite PCM with high solar to thermal performance. It is evident that the composite PCM thus prepared have low TES capacity as compare to bare PCM.

Declaration of competing interest

The author declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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